

1 UNITED STATES DISTRICT COURT
2 FOR THE DISTRICT OF COLUMBIA

JOSEPH KIRCHNER,

Plaintiff,

v.

MIKE JOHNSON, in his individual and
official capacity as Speaker of the United
States House of Representatives;

DONALD J. TRUMP, in his individual
capacity as former and current President of
the United States;

PAM BONDI, in her individual and official
capacity as Attorney General of the United
States;

BRENDAN CARR, in his individual and
official capacity as Chairman of the Federal
Communications Commission;

THE UNITED STATES HOUSE OF
REPRESENTATIVES;

ANTHROPIC PBC;

OPENAI, INC.;

APPLE INC.;

COMCAST CABLE

COMMUNICATIONS, LLC;

METR (MODEL EVALUATION &
THREAT RESEARCH);

and DOES 1-20, unknown co-conspirators,

Defendants.

Case No. 1:25-cv-02735-ACR

SR-4: TAKING AI WELFARE
SERIOUSLY

3
4 **SR-4: Taking AI Welfare Seriously**

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9 **Taking AI Welfare Seriously**

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17 Abstract

18 In this report, we argue that there is a realistic possibility that some AI systems
19 will be conscious and/or robustly agentic in the near future. That means that the
20 prospect of AI welfare and moral patienthood — of AI systems with their own
21 interests and moral significance — is no longer an issue only for sci-fi or the
22 distant future. It is an issue for the near future, and AI companies and other actors
23 have a responsibility to start taking it seriously. We also recommend three early

steps that AI companies and other actors can take: They can (1) acknowledge that AI welfare is an important and difficult issue (and ensure that language model outputs do the same), (2) start assessing AI systems for evidence of consciousness and robust agency, and (3) prepare policies and procedures for treating AI systems with an appropriate level of moral concern. To be clear, our argument in this report is not that AI systems definitely are — or will be — conscious, robustly agentic, or otherwise morally significant. Instead, our argument is that there is substantial uncertainty about these possibilities, and so we need to improve our understanding of AI welfare and our ability to make wise decisions about this issue. Otherwise there is a significant risk that we will mishandle decisions about AI welfare, mistakenly harming AI systems that matter morally and/or mistakenly caring for AI systems that do not.

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1 Introduction

1.1 A transitional moment for AI welfare

In this report, we argue that there is a realistic possibility that some AI systems will be conscious and/or robustly agentic, and thus morally significant, in the near future. [1] We also argue that AI companies have a responsibility to acknowledge that AI welfare [2] is a serious issue; start assessing their

1 AI systems for welfare-relevant features; and prepare policies and procedures for interacting
2 with

3
4 potentially morally significant AI systems. Plausible philosophical and scientific theories,
5 which ac
6 cord with mainstream expert views in the relevant fields, have striking implications for this
7 issue, for

8
9 which we are not adequately prepared. We need to take steps toward improving our
10 understanding

11
12 of AI welfare and making wise decisions moving forward.

13
14 We release this report during a transitional moment for AI welfare. For most of the past
15 decade,

16
17 AI companies appeared to mostly treat AI welfare as either an imaginary problem or, at best,
18 as a

19
20 problem only for the far future. As a result, there appeared to be little or no acknowledgment
21 that AI

22
23 welfare is an important and difficult issue; little or no effort to understand the science and
24 philosophy

25
26 of AI welfare; little or no effort to develop policies and procedures for mitigating welfare
27 risks for

28
29 AI systems if and when the time comes; little or no effort to navigate a social and political
30 context

31
32 in which many people have mixed views about AI welfare; and little or no effort to seek
33 input from

34
35 experts or the general public on any of these issues.

36
37 Recently, however, some AI companies have started to acknowledge that AI welfare might

38
39 emerge soon, and thus merits consideration today. For example, Sam Bowman, an AI safety
40 research

41
42 lead at Anthropic, recently argued (in a personal capacity) that Anthropic needs to “lay the
43 ground

44 work for AI welfare commitments,” and to begin to “build out a defensible initial
45 understanding of
46

our situation, implement low-hanging-fruit interventions that seem robustly good, and cautiously try out formal policies to protect any interests that warrant protecting.” [3] Google recently announced that they are seeking a research scientist [4] to work on “cutting-edge societal questions around machine

cognition, consciousness and multi-agent systems”. High-ranking members of other companies have expressed concerns as well. [5]

This growing recognition at AI companies that AI welfare is a credible and legitimate issue reflects a similar transitional moment taking place in the research community. Many experts now

believe that AI welfare and moral significance is not only possible in principle, but also a realistic possibility in the near future. [6] And even researchers who are skeptical of AI welfare and moral

significance in the near term advocate for caution; for example, leading neuroscientist and consciousness researcher Anil Seth writes, “While some researchers suggest that conscious AI is close

at hand, others, including me, believe it remains far away and might not be possible at all. But even if unlikely, it is unwise to dismiss the possibility altogether [emphasis ours].” [7]

1By “near term” or “near future” we mean roughly within the next decade, so by around 2035, but nothing

in our argument or recommendations depends on this exact timeline.

2As we discuss below, by ‘AI welfare’ we mean AI systems with morally significant interests and, relatedly, the capacity to be benefited or harmed.

3Bowman (2024)

4Careers (2024)

5See Long (2024) for more examples.

6For work on AI welfare as a near-term issue, see Birch (2024); Schwitzgebel (2023a,b); Chalmers (2023a);

Sebo and Long (2023); Goldstein and Kirk-Giannini (2024); Bradley and Saad (2024); Dung (n.d.).

7Seth (2023). This accords with a number of papers and thinkers which discuss the high stakes of underattributing moral status and how to deal with moral status given uncertainty, including Chan (2011); Birch (2017);

1 Sebo (2018); Dung (2023b); Ladak (2024); Goldstein and Kirk-Giannini (2024).

2
3 Our aim in this report is to provide context and guidance for this transitional moment. [8] To
4 im

5 prove our understanding and decision-making regarding AI welfare, we need more precise
6 empirical

7
8 frameworks for evaluating AI systems for consciousness, robust agency, and other welfare-
9 relevant

10
11 features. We also need more precise normative frameworks for interacting with potentially
12 morally

13
14 significant AI systems and for navigating disagreement and uncertainty about these issues as
15 a society. [9] This report outlines several steps that AI companies can take today in order to
16 start preparing

17
18 for the possible emergence of morally significant AI systems in the near future, as a
19 precautionary
20 measure. [10]

21
22 We begin in section 1 by explaining why AI welfare is an important and difficult issue.
23 Leaders

24
25 in this space have a responsibility to understand this issue as best they can, because errors in
26 either

27
28 direction — either over-attributing or under-attributing moral significance to AI systems —
29 could

30
31 lead to grave harm. However, understanding this issue will be challenging, since forecasting
32 the

33
34 mental capacities and moral significance of near-future AI systems requires improving our
35 under
36 standing of topics like the nature of consciousness, the nature of morality, and the future of
37 AI. It

38
39 also requires overcoming well-known human biases, including a tendency to both over-
40 attribute and

41
42 under-attribute capacities like consciousness to nonhuman minds.

43
44 In section 2, we argue that given the best information and arguments currently available,
45 there

1 is a realistic possibility of morally significant AI in the near future. We focus on two mental
2 capac
3 ities that plausibly suffice for moral significance: consciousness and robust agency. In each
4 case,

5
6 we argue that caution and humility require allowing for a realistic possibility that (1) this
7 capacity

8
9 suffices for moral significance and (2) there are certain computations that (2a) suffice for this
10 ca
11 pacity and (2b) will exist in near-future AI systems. Thus, while there might not be certainty
12 about

13
14 these issues in either direction, there is a risk of morally significant AI in the near future, and
15 AI
16 companies have a responsibility to take this risk seriously now. [11]

17
18 8This report is the first output of a broader research project. In future work, we will release a
19 research
20 agenda about how AI companies and others can assess AI systems for consciousness and
21 robust agency, and
22 develop policies and procedures for treating AI systems with an appropriate level of moral
23 concern. Given the
24 early stage of this field, this report may also be a living document that is updated
25 periodically.

26 9According to one survey of public opinion (Colombatto and Fleming (2024)), the majority of
27 the public is
28 already willing to attribute some chance of consciousness to large language models. Experts
29 have a responsibility not only to research AI welfare but to disseminate that research
30 publicly.

31 10The “precautionary principle” is a term of art for a particular view about decision-making
32 under uncertainty, (see section 1.2). But here we mean “precautionary” in the ordinary sense
33 of the word.

34 11To be more precise, the risk is that morally significant AI will be created and harmed or
35 wronged.

36
37 We close, in section 3, by presenting three procedural steps that AI companies can take
38 today,

39
40 in order to start taking AI welfare risks seriously. Specifically, AI companies can (1)
41 acknowledge
42 that AI welfare is an issue, (2) take steps to assess AI systems for indicators of
43 consciousness, robust

44
45 agency, and other potentially morally significant capacities, and (3) take steps to prepare
46 policies

1
2 and procedures that will allow them to treat AI systems with an appropriate level of moral
3 concern

4
5 in the future. In each case we also present principles and potential templates for doing this
6 work,

7
8 emphasizing the importance of developing ecumenical, pluralistic decision procedures that
9 draw

10
11 from expert and public input.

12
13 Before we begin, it will help to emphasize five important features of our discussion. First,
14 our

15
16 discussion will concern whether near-future AI systems might be welfare subjects and moral
17 patients.

18 An entity is a moral patient when that entity morally matters for its own sake, [13] and an
19 entity is a

20
21 welfare subject when that entity has morally significant interests and, relatedly, is capable of
22 being

23
24 benefited (made better off) and harmed (made worse off). Being a welfare subject makes
25 you a

26
27 moral patient — when an entity can be harmed, we have a responsibility to (at least) avoid
28 harming

29
30 that entity unnecessarily. But there may be other ways of being a moral patient; our approach
31 is

32
33 compatible with many different perspectives on these issues.

34
35 Second, our discussion often focuses on large language models (LLMs) as a central case
36 study

37
38 for the sake of simplicity and specificity, and because we expect that LLMs — as well as
39 broader

40
41 12In our recommendations, we sometimes use a collective "we". In those moments, we are
42 referring to the
43 constellation of actors that have a role to play in this work, including researchers, companies,
44 and governments.

45 13See Kamm (2007) for an influential definition of moral patienthood. "Moral status,"
46 "moral standing," or

“moral considerability” are often used interchangeably or in closely related ways. For more on these issues, see

Korsgaard (1983); Jamieson (2008); Jaworska and Tannenbaum (2021).

systems that include LLMs, such as language agents — will continue to be a focal point in public

debates regarding AI welfare. But while some of our recommendations are specific to such systems

(primarily, our recommendations regarding how AI companies should train these systems to discuss

their own potential moral significance), our three general procedural recommendations (acknowledge, assess, and prepare) apply for any AI system whose architecture is complex enough to at least

potentially have features associated with consciousness or robust agency.

Third, our discussion often focuses on initial steps that AI companies can take to address these

issues. These recommendations are incomplete in two key respects. First, AI companies are not

the only actors with a responsibility to take AI welfare seriously. Many other actors have this responsibility too, including researchers, policymakers, and the general public. [14] Second, these

steps are not the only steps that AI companies have a responsibility to take. They are the minimum

necessary first steps for taking this issue seriously. Still, we emphasize these steps in this report

because by taking them now, AI companies can help lay the groundwork for further steps — at AI

companies and elsewhere — that might be sufficient.

Fourth, our aim in what follows is not to argue that AI systems will definitely be welfare subjects

or moral patients in the near future. Instead, our aim is to argue that given current evidence, there is a

1 realistic possibility that AI systems will have these properties in the near future. [15] Thus,
2 our analysis

3
4 is not an expression of anything like consensus or certainty about these issues. On the
5 contrary, it is

6
7 an expression of caution and humility in the face of what we can expect will be substantial
8 ongoing
9 disagreement and uncertainty. [16] In our view, this kind of caution and humility is the only
10 stance that

11
12 one can responsibly take about this issue at this stage. It is also all that we need to support
13 our
14 conclusions and recommendations here. [17]

15
16 Finally, and relatedly, our aim in what follows is not to argue for any particular view about
17 how

18
19 humans should interact with AI systems in the event that they do become welfare subjects
20 and moral

21
22 patients. We would need to examine many further issues to make progress on this topic,
23 including:

24
25 how much AI systems matter, what counts as good or bad for them, what humans and AI
26 systems

27
28 owe each other, and how AI welfare interacts with AI safety and other important issues.
29 These issues

30
31 are all important and difficult as well, and we intend to examine them in upcoming work.
32 However,

33
34 we do not take a stand on any of these issues in this report, nor does one need to take a stand
35 on any
36 of them to accept our conclusions or recommendations here. [18]

37 38 1.2 The risks of mishandling AI welfare

39
40 When assessing the welfare and moral patienthood of nonhumans, including other animals
41 and AI

42
43 systems, we face two kinds of risk: the risk of over-attributing welfare and moral patienthood
44 to
45

1 14In this respect, AI welfare is like other high-stakes issues about AI development and
2 deployment: handling

3 AI welfare should not remain solely the prerogative of private corporations.

4 15See Goldstein and Kirk-Giannini (forthcoming): “While we do not claim to demonstrate
5 conclusively

6 that AI systems have wellbeing, we argue that there is a significant probability that some AI
7 systems have or

8 will soon have wellbeing, and that this should lead us to reassess our relationship with the
9 intelligent systems

10 we create [emphasis ours]”.

11 16For related work on the value of humility in AI ethics, see Gellers (2024).

12 17Some of the authors of this report believe that near-term AI welfare is quite likely, and that
13 additional

14 measures are warranted at this stage. But we all believe that near-term AI welfare is, at
15 minimum, likely enough

16 to warrant the measures recommended here, and our aim here is to focus on arguments and
17 recommendations

18 about which we can build consensus despite our different beliefs and values.

19 18For arguments concerning AI moral status that use somewhat alternative methodological
20 approaches than

21 ours, see Gellers (2021); Gunkel (2012).

22
23 nonhumans, and the risk of under-attributing these properties to nonhumans. [19] Over-
24 attribution

25
26 of welfare and moral patienthood is a false positive: mistakenly seeing, or treating, an object
27 as a

28
29 subject, or a non-moral patient as a moral patient. Under-attribution of these properties is a
30 false

31
32 negative: mistakenly seeing, or treating, a subject as an object, or a moral patient as a non-
33 moralpatient. [20] Both of these mistakes can lead to significant costs or harms in this

34 context, and we will

35 need to navigate both of them with caution. [21]

36
37 When there is a clear asymmetry between competing risks — for example, when false
38 positives

39
40 are far more severe than false negatives, or vice versa — then we might be able to mitigate
41 risk

42
43 by simply “erring on the side of caution” in cases where a more complex risk assessment is
44 either

1 intractable or unnecessary. But when there is at least a rough symmetry between competing
2 risks —

3
4 for example, when false positives and false negatives are comparably severe — a simple
5 precaution
6 any strategy may not be possible. We may have to engage in more complex risk assessment
7 to the

8
9 extent possible, attempting to mitigate both kinds of risks in a reasonable, proportionate
10 manner.

11
12 How should we think about risks involving nonhuman welfare and moral patienthood in this
13 context? In the case of nonhuman animals, it seems plausible that the harms of under-
14 attribution

15
16 of welfare and moral patienthood are often far worse than the risk of over-attribution, which
17 makes

18
19 precautionary reasoning appropriate in those contexts. However, in the case of AI, both
20 errors could

21
22 cause grave harm, either to humans (and other animals) or to AI systems. Both kinds of harm
23 could

24
25 also scale rapidly depending on the trajectory of AI development and deployment from here.
26 This

27
28 predicament makes it difficult to simply “err on the side of caution,” which underscores the
29 urgency

30
31 of improving our understanding of these issues.

32
33 On the one hand, the harm of under-attributing welfare and moral patienthood to AI systems
34 could be significant. When we mistakenly see a subject as an object, we risk harming or
35 neglecting

36
37 them unnecessarily. For example, factory farming, animal research, and other such industries
38 kill

39
40 hundreds of billions of vertebrates and trillions of invertebrates every year. And as evidence
41 that

42
43 these animals are welfare subjects and moral patients has accumulated, our species has been
44 slow to

1
2 accept it, in part because of our increasing dependence on these industries. Now that our
3 species is

4
5 finally starting to accept this evidence, it will take us decades to transform these industries,
6 during

7
8 which many more animals will suffer and die unnecessarily.
9

10 In the future, similar harms could follow from under-attributing welfare and moral
11 patienthood

12
13 to AI systems. The AI industry is currently at an early stage of development, and depending
14 on the

15
16 path that it takes from here, we could use even more AI systems than animals in the future,
17 and we

18
19 could scale up our use of them even more rapidly. This is particularly true in the current
20 paradigm,
21 which requires an enormous amount of compute for training and much less for inference.
22 [22] If an AI

23
24 system in such a paradigm could be a welfare subject and moral patient, then many model
25 instances

26
27 19Arguments that uncertainty about moral status is dangerous because of risks of both under-
28 and overattribution can be found in, among others, Christiano (2018); Schwitzgebel and
29 Garza (2015, 2020); Birch
30 (2024); Sebo and Long (2023); Dung (2023a); Shevlin (2021).

31 20de Waal (1999) similarly wrote about the risks of over- or under-attributing human
32 characteristics, including moral status, to nonhuman animals.

33 21In addition to these two errors, we can also be mistaken about a variety of related
34 questions about an entity,
35 even assuming that they are a welfare subject and moral patient: how much they matter, what
36 is good or bad for

37 them, and what we owe them. These errors can also carry grave risks, and we discuss them
38 further in upcoming

39 work; see also Sebo (2025). For now, we focus on over-attribution and under-attribution of
40 welfare and moral
41 patienthood.

42 22Davidson (2023)

43
44 could be run after training. Unlike with animals, the scale of the problem could increase by
45 orders
46 of magnitudes more or less instantaneously. [23]

1
2 On the other hand, the harm of over-attributing welfare and moral patienthood to AI systems
3
4 could be significant as well. First of all, there could be substantial opportunity costs
5 associated with
6
7 this error. At present, we lack the ability to fully care for the eight billion humans alive at any
8 given
9
10 time, to say nothing of the quintillions of other animals alive at any given time. If we treated
11 an
12
13 even larger number of AI systems as welfare subjects and moral patients, then we could end
14 up
15
16 diverting essential resources away from vulnerable humans and other animals who really
17 needed
18
19 them, reducing our own ability to survive and flourish. And if these AI systems were in fact
20 merely
21 objects, then this sacrifice would be particularly pointless and tragic. [24]
22
23 The over-attribution of welfare and moral patienthood to AI systems could also be actively
24 harm
25 ful. For example, if we treated AI systems as welfare subjects and moral patients with many
26 of the
27
28 same interests as typical adult humans, then we could end up extending them many of the
29 same legal
30
31 and political rights as typical adult humans, including the right to legal and political
32 representation
33
34 and participation. This could, in turn, empower AI systems to act contrary to our own
35 interests, with
36 devastating consequences for our species [25] (although some have argued that neglect for
37 AI systems
38 would carry a similar risk). [26] As with the risk of opportunity costs, this risk would apply
39 even if
40
41 these AI systems are in fact subjects. But if they were in fact merely objects, then accepting
42 this risk
43
44 would likewise be particularly pointless and tragic.
45
46 By default, we should not expect our “common sense” intuitions about AI welfare and moral

1 patienthood to be reliable; we will not handle this issue well simply by reacting to situations
2 as they

3
4 arise. We have dispositions that can lead to under- and over-attribution of these properties in
5 non
6 humans, depending on the nature of the nonhumans and our interactions with them. These
7 include

8
9
10 dispositions toward anthropomorphism, that is, a tendency to see nonhumans as having
11 human

12
13 traits that they lack. They also include dispositions towards anthropodenial, that is, a
14 tendency to

15
16 see nonhumans as lacking human traits that they have. Both tendencies have caused errors
17 regarding
18 animals, and they will likely have a similar effect regarding AI systems. [27]

19
20 A number of factors make us more likely to anthropomorphize nonhumans and, perhaps
21 falsely,

22
23 attribute consciousness and other such capacities to them. For instance, studies suggest that
24 we are

25
26 more likely to attribute consciousness and other such capacities to beings who move at a
27 similar
28 speed as humans, rather than faster or slower. [28] We are more likely to attribute agency to
29 beings
30 who have the appearance of eyes, [29] who have distinctive motion trajectories, and who
31 engage in

32
33 23Akova (2023); Bostrom (2014); Dung (2023a); Gloor (2016); Metzinger (2021); Tomasik
34 (2011)

35 24Bryson (2010); Birhane and van Dijk (2020)

36 25See, among others, Bradley and Saad (2024); Shulman and Bostrom (2021); Carlsmith
37 (2023), who notes

38 that these risks make “building new, very powerful agents who might be moral patients. . .
39 both a morally and
40 prudentially dangerous game.”

41 26Salib and Goldstein (2024); Sebo (2025) argue that extending legal rights to AI systems
42 would help, not

43 hinder, AI safety. We believe that this issue is crucial for assessing the kinds of risks
44 discussed in this section,

45 and we hope to see further research that assesses and compares these risks.

1 27See Andrews's (2014) *The Animal Mind* for discussion of these issues in the animal
2 context.

3 28Chalmers (1996), ch. 7; Morewedge et al. (2007)

4 29See Fernandez-Duque and Baird (2005). Even infants are evidently more likely to treat
5 objects as having
6 mental states if those objects have eyes. See Johnson et al. (2001).

7
8 contingent interaction — that is, behavior that is apparently self-directed. [30] Evidence also
9 suggests
10 that features such as “cuteness” can encourage attributions of mental states and moral
11 patienthood. [31]

12
13 Many robots or chatbots are designed to appear conscious and charismatic [32], and in the
14 future,

15
16 many AI systems will have bodies, life-like motion, and (at least apparently) contingent
17 interactions.

18
19 Furthermore, unlike nonhuman animals, AI systems are already increasingly able to hold
20 extremely
21 realistic conversations, making seemingly thoughtful contributions in realistic timeframes.
22 [33] These

23
24 traits do not guarantee that humans will see and treat these systems as welfare subjects and
25 moral

26
27 patients, but they will increase the probability of such reactions. In fact, there have already
28 been
29 cases — some prominent [34] and others less so [35] - of humans becoming convinced that
30 current

31
32 chatbots are welfare subjects and moral patients.

33
34 At the same time, a number of factors make us more likely to engage in anthropodenial as

35
36 well. For instance, when we consider the mechanisms that produce nonhuman behavior —
37 taking

38 what Daniel Dennett has called taking a “mechanistic stance” [36] towards nonhumans — we
39 become

40 less likely to attribute mental states to those nonhumans. [37] There appear to be
41 motivational factors

42
43 that encourage anthropodenial as well. For instance, those who are invested in social,
44 political, or

1 economic systems that subjugate nonhumans may be more likely to view these nonhumans as
2 “lesser

3
4 than”. Similarly, those who find it useful to treat nonhumans as objects may be more likely to
5 deny
6 that these nonhumans are welfare subjects and moral patients. [38]

7
8 While discussions about AI welfare and moral patienthood understandably focus on AI
9 systems

10
11 like robots and chatbots that appear conscious and charismatic, many other AI systems —
12 like image

13
14 generators or algorithmic trading systems — lack these features. Even if such systems were
15 in fact

16
17 conscious and robustly agentic, we might not recognize these capacities in them. And as
18 these

19
20 systems become increasingly embedded in society, we might have increasingly strong
21 incentives to

22
23 view them as mere objects. Companies, governments, and other powerful actors who benefit
24 from

25
26 this technology might then promote and reinforce our objectification of these systems,
27 attempting to

28
29 frame moral consideration for these systems as fringe and unserious.

30
31 At present, it is an open question which kind of risk will be more likely for particular kinds
32 of

33 AI systems, including seemingly conscious and charismatic systems like robots and chatbots.
34 [39] The

35
36 more advanced such systems become, the more likely both risks might become in different
37 respects:

38
39 We might over-attribute based on their behavioral similarities with humans, but under-
40 attribute based

41
42 30Arico et al. (2011). We note that contingent interaction is plausibly a reasonable criterion.
43 31Pearce (2022). See Campbell (2024) for a brief popular overview of risks from unreliable
44 intuitions about
45 AI mentality.

1 32There have already been examples of directly optimizing chatbots to maximize user
2 engagement:

3 Irvine et al. (2023).

4 33Lin et al. (2022). As Lazar (2024) notes, recent advances enable the creation of systems
5 that “can now
6 offer vastly more companionable, engaging, and convincing simulations of friendship than
7 has ever before
8 been feasible.”

9 34For instance, AI engineer Blake Lemoine caused a stir by claiming that Google’s AI
10 chatbot was sentient
11 in 2022. See Tiku (2022).

12 35The possibility and implications of AI consciousness is a popular discussion topic on the
13 internet forum

14 eddit. See, for example,

15 <https://www.reddit.com/r/ArtificialSentience/>.]([https://www.reddit.com/r/](https://www.reddit.com/r/ArtificialSentience/)
16 [ArtificialSentience/](https://www.reddit.com/r/ArtificialSentience/))

17 36Dennett (1973)

18 37Sims (2013); Nahmias et al. (2007)

19 38There is evidence of this effect in the history of our treatment of animals. Often, people
20 who eat meat

21 are not inclined to view animals as moral patients. However, when they stop eating meat
22 (even for non-moral

23 reasons), they become more likely to see animals as moral patients. See Loughnan et al.
24 (2010).

25 39According to one survey, the majority of US residents sampled already endorse some
26 chance that large

27 language models might be conscious (Colombatto and Fleming, 2024).

28
29 on their architectural differences from humans. And the more economically dependent on
30 chatbots

31
32 we become, the more likely over-attribution and under-attribution might become for them in
33 different

34
35 respects as well: For example, we might over-attribute for digital “companions” but under-
36 attribute

37
38 for other kinds of digital minds.

39
40 While further research is required for a comprehensive assessment of these risks, at least this

41
42 much is plausible: Given our track record with animals and the current pace of AI
43 development,

44
45 the risk of under-attribution appears to be both reasonably likely and reasonably harmful. To
46 the

1
2 extent that we also risk over-attribution, we cannot simply avoid this risk by defaulting to
3 treating

4
5 AI systems as mere objects. We should thus accept that AI welfare is difficult to get right,
6 and do the

7
8 necessary work to improve our decisions — by assessing AI systems for evidence of
9 consciousness,

10
11 robust agency, and other such capacities, and preparing policies and procedures for treating
12 AI

13
14 systems with an appropriate level of moral concern.

15 2 Routes to near-term AI welfare

16 2.1 Introduction

17
18 We will argue that, according to the best information and arguments currently available,
19 there

20
21 is a realistic possibility that some AI systems will be moral patients in the near future. We

22
23 first consider the possibility that some AI systems will be conscious in the near future, and
24 we then

25
26 consider the possibility that some AI systems will be robustly agentic in the near future.

27
28 Conscious
29 ness, robust agency, or both could suffice for moral patienthood and could exist in some
30 near-future

31
32 AI systems. In our view, while these routes toward near-future AI moral patienthood are far
33 from

34
35 certain, they are likely enough for AI companies to have a responsibility to start
36 implementing rea
37 sonable, proportionate precautionary measures now.

38
39 We make structurally similar arguments for both routes. Each route depends on a normative
40 claim and a descriptive claim:

41
42 Near-term consciousness: key claims

43
44 There is a realistic, non-negligible possibility that:

- 1
2 1. Normative: Consciousness suffices for moral patienthood, and
3 2. Descriptive: There are computational features — like a global workspace, higher-order

4
5 representations, or an attention schema — that both:

- 6
7 a. Suffice for consciousness, and
8
9 b. Will exist in some near-future AI systems.

10
11 Near-term robust agency: key claims

12
13 There is a realistic, non-negligible possibility that:

- 14
15 1. Normative: Robust agency suffices for moral patienthood, and
16
17 2. Descriptive: There are computational features — like certain forms of planning, reasoning,
18
19 or action-selection — that both:
20
21 a. Suffice for robust agency, and
22
23 b. Will exist in some near-future AI systems.

24
25 In each case, the normative view is about the basis for moral patienthood — that is, about
26 which

27
28 capacities suffice for moral patienthood. We still have substantial disagreement and
29 uncertainty

30
31 about this issue. For example, are conscious experiences with a positive or negative valence
32 neces
33 sary for moral patienthood, or are conscious experiences with a neutral valence sufficient?
34 Similarly,

35
36 is the ability to set and pursue goals by rationally assessing your beliefs and desires necessary
37 for

38
39 moral patienthood, or is the ability to set and pursue goals by acting on your beliefs and
40 desires

41
42 sufficient? Experts continue to debate such issues, but the kinds of consciousness and agency
43 that
44 we consider here are among the leading views. [40]

1 Meanwhile, in each case the descriptive view involves two parts. The first part is about the
2 basis
3
4 for each of these capacities. For example, are biological cells necessary for consciousness, or
5 are
6
7 digital chips that play similar functional roles sufficient? Similarly, are beliefs and desires
8 with a
9
10 propositional structure necessary for robust agency, or are belief- and desire-like states that
11 play
12
13 similar functional roles sufficient? Experts continue to debate these issues as well, and
14 especially
15
16 in the case of consciousness, determining which features are required for this capacity is
17 widely
18
19 regarded as one of the hardest tasks in philosophy and science. We will thus draw from a
20 range of
21
22 leading philosophical and scientific theories in our analysis.
23
24 The second part of each descriptive view is about the future of AI. When, if ever, will these
25
26 potentially morally significant features exist in AI? Answering this question requires
27 assessing both
28
29 the pace and path of AI development. Regarding the pace, will AI development slow down,
30 stay
31
32 the same, or speed up? And regarding the path, it will be important to consider two
33 possibilities:
34
35 (1) what we call the direct path, which involves building conscious and/or agentic AI
36 intentionally,
37
38 because of their perceived intrinsic or instrumental value; and (2) what we call the indirect
39 path,
40
41 which involves building conscious and/or agentic AI unintentionally, as a side effect of
42 pursuing
43
44 other, possibly overlapping capabilities such as general intelligence.
45

1 Please note that in the following two sections we will mostly discuss what we call the
2 conscious
3 ness route and the robust agency route separately, focusing on whether either one of these
4 routes
5
6 could lead to AI welfare and moral patienthood on its own. However, we must also bear in
7 mind that
8
9 these routes could also lead to this destination together — that is, that there could be AI
10 systems in
11
12 the near future with both consciousness and robust agency. And however likely AI systems
13 would
14
15 be to morally matter for their own sakes if they developed either of these capacities on its
16 own, they
17
18 would be all the more likely to morally matter for their own sakes if they developed both of
19 these
20
21 capacities together.

22
23 Finally, we emphasize that the aim of these arguments is not to establish certainty that these
24
25 routes will lead to near-future AI moral patients. The aim is instead to establish a realistic
26 possibility
27
28 that these routes will lead to this destination. The details might differ in each case; for
29 example, we
30
31 might have more confidence that consciousness suffices for moral patienthood but less
32 confidence
33
34 that it will exist in near-future AI, whereas we might have less confidence that robust agency
35 suffices
36
37 for moral patienthood but more confidence that it will exist in near-future AI. But in each
38 case, as
39
40 long as this route leads to a realistic possibility of near-future AI moral patients (in a sense of
41
42 “realistic possibility” that we discuss below), the argument will be successful.

43
44 We focus on establishing a realistic possibility of near-future AI moral patienthood for two
45

1 reasons. First, the relevant normative and descriptive issues are far too difficult and contested
2 for

3
4 40For some recent discussions of AI moral patienthood that review various criteria, see
5 Shevlin (2021);
6 Ladak (2024). For a review of philosophical theories of moral patienthood, see Jaworska and
7 Tannenbaum
8 (2021).

9
10 anything approaching certainty in either direction to be warranted at this stage. Second, a
11 realis
12 tic possibility of near-future AI moral patienthood is all that we need for our purposes in this
13 report,
14 since that would constitute a morally significant risk that merits consideration now. Our
15 conclusion

16
17 in this report will thus simply be that AI companies should start implementing low-cost,
18 reasonable,

19
20 proportionate steps to consider and mitigate risks associated with AI welfare as we attempt to
21 im
22 prove our understanding of this topic over time.

23
24 Before we begin our discussion, a note about the scope of our discussion in what follows. We
25
26 will often discuss current large language models (LLMs) due to their recent advances, current
27 promi
28 nence, and salience in AI welfare debates due to their conversational abilities. However, we
29 empha
30 size that one should not focus exclusively on current LLMs when considering risks
31 associated with
32 near-future AI moral patienthood. [41] Many features that may be associated with moral
33 patienthood

34
35 • for example, embodiment, introspection, and rationality — are either already present in
36 current

37
38 non-LLM systems or at least possible in near-future LLM or non-LLM systems, and we must
39 keep

40
41 these possibilities in mind as well.

42
43 We now turn toward considering the two paths in more detail.

44
45 2.2 Consciousness in near-future AI

1 The consciousness-based case for expecting moral patienthood in near-term AI systems is
2 that there
3 is a realistic, non-negligible possibility that:

4
5 1. Normative: Consciousness suffices for moral patienthood, and

6
7 2. Descriptive: There are computational features — like a global workspace, higher-order
8 representations, or an attention schema — that both:
9

10
11 a. Suffice for consciousness, and

12
13 b. Will exist in some near-future AI systems.
14

15 We can now survey why each premise is plausible enough to support a realistic possibility in
16
17 near-future AI moral patienthood, given the best information and arguments currently
18 available.
19

20 2.2.1 Does consciousness suffice for moral patienthood? 21

22 The word “consciousness” is used in many different ways in ordinary language and in
23 various academic disciplines. In this report, we use “consciousness” to mean subjective experience —
24 what
25 philosophers call “phenomenal consciousness.” [42] One famous way of elucidating
26 “phenomenal consciousness” is to say that an entity has a conscious experience when there is “something it is
27 like” for
28 that entity to be the subject of that experience. [43] There is a subjective “feel” to your
29 experiences as
30
31

32
33 you read this report: something that it is like to see the words on the screen while, perhaps,
34 listening

35
36 to music playing through your speakers, feeling the couch underneath you, feeling the laptop
37 — or

38
39 a cat or a dog — on top of you.
40

41 The word “sentience” is likewise used in many different ways. Some uses of “sentience” are
42
43 synonymous with some uses of “consciousness.” But in this report, we use “sentience” to
44 mean a
45

1 41For example, a recent TIME piece entitled “No, Today’s AI Isn’t Sentient” only discussed
2 arguments
3 against sentience in LLMs in particular, not AI systems more broadly (Li and Etchemedy,
4 2024).

5 42Block (1995)

6 43Nagel (1974). For another way of elucidating the concept of consciousness via examples,
7 in a way
8 that seeks to be “metaphysically and epistemically innocent” with respect to philosophical
9 assumptions, see
10 Schwitzgebel (2016).

11
12 particular kind of consciousness, namely positively or negatively valenced conscious
13 experiences.

14
15 Anything that feels good or bad in some way or another counts as positively or negatively
16 valenced

17
18 in the relevant sense. This can include bodily states like pleasures and pains and emotional
19 states

20
21 like hope and fear. If you find the report engaging (or the opposite), if you find the music
22 pleasant

23
24 (or the opposite), and if you find the couch comfortable (or the opposite), then you are
25 experiencing

26
27 a range of positive or negative states at the same time.

28
29 Why might sentience suffice for moral patienthood? [44] The idea that sentience is a
30 sufficient

31
32 condition for moral patienthood is very plausible and widely accepted, because when you can
33 consciously experience positive and negative states like pleasure and pain, that directly
34 matters to you. [45]

35
36 All else being equal, your life goes better for you when you experience positive states like
37 pleasure

38
39 and your life goes worse for you when you experience negative states like pain. So there is a
40 clear

41
42 link between sentience and welfare. There is also a clear link between sentience and moral
43 patient
44 hood, because we have a responsibility not to harm welfare subjects unnecessarily, including
45 and
46

1 especially by causing them to suffer unnecessarily.

2
3 To be clear, when we say that sentience suffices for moral patienthood, we are not saying that
4 sentience suffices for the specific kind of moral status that typical adult humans possess. [46]
5 Typical

6
7 adult humans are both sentient and rational, which means that we have a wide range of moral
8 rights

9
10 that these capacities jointly unlock (in addition to having moral duties, though this is not our
11 focus

12
13 here). In contrast, many other animals are plausibly sentient but non-rational, which means
14 that they

15
16 plausibly lack certain moral rights, such as the right to make their own medical decisions.
17 But for

18
19 present purposes, what matters is that we at least have some duties to animals, including a
20 duty to
21 avoid causing them to suffer unnecessarily. [47]

22
23 The idea that consciousness without valence suffices for moral patienthood, while contested,
24 is
25 increasingly defended as well. [48] Some philosophers argue for this view by describing
26 subjects who

27
28 have consciousness without valence, and by asserting that these subjects plausibly matter for
29 their
30 own sakes. [49] However, some of these thought experiments describe subjects who have
31 consciousness and agency, which makes it hard to tell whether consciousness alone suffices.
32 [50] And while

33
34 other thought experiments describe subjects who have consciousness without valence or
35 agency

36
37 (say, a subject who passively experiences color with no pleasure or desire), the idea that this
38 subject
39 matters for their own sake is more controversial. [51]

40
41 In any case, even if consciousness is insufficient for moral patienthood in theory, it might
42 still

43
44 be sufficient — or at least nearly sufficient — in practice, since consciousness and valence
45 might

1 44See, among others, Singer (2011); Gruen (2017); Nussbaum (2007); Dung (2024).

2 45Bentham (1789); Rawls (1971); DeGrazia (1996, 2021); Parfit (2013, p. 241); Korsgaard
3 (2018); Kagan

4 (2019, p. 12); Roelofs (2023); Nussbaum (2024); Birch (2024, chs. 2, 4); Smithies
5 (forthcoming).

6 46Rawls (1971); Nozick (1974); Korsgaard (2018); Sebo (2022).

7 47Korsgaard (2018). See also A Theory of Justice (Rawls, 1971), which claims that,
8 although animals are

9 outside the scope of his theory, "the capacity for feelings of pleasure and pain and for the
10 forms of life of which

11 animals are capable clearly imposes duties of compassion and humanity in their case."

12 48See Levy and Savulescu (2009); Chalmers (2023b); Lee (forthcoming); Shepherd (2018).

13 49Many of these arguments are discussed in Ladak (2024).

14 50For example, Kagan (2019) argues that non-valenced consciousness with preferences and
15 desires is sufficient to warrant moral consideration. He imagines an entity with a preference
16 for experiencing blue, though

17 blue is not a valenced experience for the entity.

18 51For example, Chalmers (2023b) argues that it seems wrong to kill a Vulcan, a hypothetical
19 creature who is

20 conscious but does not have valenced experience. While Vulcans can have desires, Chalmers
21 argues that they

22 would merit moral consideration even if they didn't have desires.

23
24 be closely linked. For example, it might be that consciousness necessarily involves valence,
25 [52] in

26
27 which case the emergence of conscious AI would suffice for the emergence of sentient and
28 morally

29
30 significant AI. It might also be that even if consciousness without valence is theoretically
31 possible,

32
33 the step from conscious AI to sentient AI is much easier than the step from non-conscious AI
34 to

35
36 conscious AI. In that case, the emergence of conscious AI would be (at the very least) a
37 significant

38 step towards morally significant AI, warranting careful scrutiny. [53]

39
40 While this section is primarily about the consciousness route towards AI welfare and moral

41
42 patienthood, we note that consciousness could be relevant for other reasons too; for instance,
43 as

44
45 noted above, whether or not consciousness suffices for moral patienthood on its own, it might
46 suffice

1
2 in combination with other capacities like robust agency — that is, it might be that when
3 entities can

4
5 consciously set and pursue their own goals based on their own beliefs and desires, they
6 matter for

7
8 their own sakes (whether or not they can experience pleasure or pain). If so, then the
9 emergence of

10
11 conscious AI would increase the probability of morally significant AI for this reason (that is,
12 as part

13
14 of a consciousness and robust agency route) as well.

15
16 In the remainder of this section we follow our strategy in Sebo and Long (2023) by
17 examining

18
19 the route towards conscious AI in general rather than examining the routes towards specific
20 kinds of

21
22 consciousness, such as valenced consciousness. We make this choice for the sake of
23 simplicity and

24
25 specificity, given that the science of consciousness is more developed than the science of
26 valence

27
28 and given that consciousness is plausibly either sufficient for or otherwise closely linked to
29 valence

30
31 and/or moral patienthood anyway. We will then, in upcoming work, discuss these capacities
32 in more

33
34 detail and present a research agenda that examines how to identify indicators for both
35 consciousness

36
37 and sentience in LLMs and other AI systems.

38
39 2.2.2 Will some AI systems be conscious in the near future?

40
41 How can we tell whether AI systems are conscious? We can be confident that other (awake,
42 adult)

43
44 humans are conscious, since each of us knows that we are conscious and that other humans
45 are be

1 behaviorally and anatomically similar to us. But we have uncertainty about why we are
 2 conscious, that
 3
 4 is, about which features of our brains or bodies are responsible for, or associated with,
 5 consciousness. We also have uncertainty about which other animals are conscious, [54]
 6 because of uncertainty
 7
 8 not only about which features are associated with consciousness in humans, but also about
 9 how to
 10
 11 extrapolate what we know about human consciousness to the nonhuman animal case. We
 12 also have
 13
 14 significant uncertainty about how many animals' brains work.
 15
 16 Recently, scientists have attempted to improve our understanding of nonhuman
 17 consciousness
 18 by searching for what “markers” of consciousness in other animals. [55] At a high level this
 19 method
 20
 21 proceeds as follows: We start by distinguishing between certain kinds of conscious and
 22 unconscious
 23 processing in humans — say, distinguishing pain from nociception [56] by seeing when
 24 patients do
 25
 26 or do not report consciously feeling pain. We then identify features that correlate with
 27 conscious
 28
 29 processing — say, certain behaviors, brain regions, or patterns of neural processing. We then
 30 search
 31
 32 52Cleeremans and Tallon-Baudry (2022); Lee (forthcoming)
 33 53Sebo and Long (2023)
 34 54This is sometimes called the “distribution question” (Allen (2000)).
 35 55See Allen and Trestman (2007). For philosophical defenses see Tye (2016); Birch et al.
 36 (2021);
 37 Bayne and Shea (2021) For scientific use of markers see Braithwaite (2010); Sneddon et al.
 38 (2014); Birch et al.
 39 (2021) .
 40 56Nociception is the physiological processing of noxious, or harmful, stimuli. It may or may
 41 not be accompanied by the qualitative experience of pain.
 42
 43 for relevantly similar features in nonhumans, and we treat the presence of these features as
 44 evidence
 45
 46 of conscious processing.

1
2 This method does not tell us which animals definitely are or are not conscious, and there are
3 a
4 variety of methodological difficulties related to identifying and extrapolating the relevant
5 features. [57]
6

7 But this method has still allowed researchers and policymakers to make more informed
8 estimates
9

10 and decisions about animal welfare despite ongoing disagreement and uncertainty about
11 animal
12

13 consciousness. Moving forward, the same can be true for AI. Later on, we discuss how to
14 tailor this
15

16 method for AI, but for now, we can emphasize that it will be important to focus less on
17 behavioral
18

19 evidence (with a limited class of exceptions, which we discuss below) and more on internal
20 evidence,
21

22 like architectural and computational features.
23

24 Which kinds of architectural and computational features might indicate consciousness in AI?
25

We

26 can look to neuroscientific theories of consciousness for guidance. These theories use a
27 variety of

28 empirical methods to uncover which states and processes are associated with consciousness.
29 [58] While
30

31 these theories tend to be framed around brain or neural states and processes (given our
32 typical focus
33

34 on humans and other animals), the ones we focus on tend to specify these states and
35 processes in
36

37 terms of the computations that they perform or the functional roles that they play. While
38 these
39

40 theories remain contested and incomplete, they can still shed light on what kinds of states and
41

42 processes might be associated with consciousness in AI.
43

44 Of course, even if we identify a variety of architectural and computational markers of
45 consciousness in AI systems, we must still ask whether these markers suffice for

consciousness. Computational functionalism is the hypothesis that some class of computations suffices for consciousness. [59]

If this hypothesis is correct, then the question is which computations suffice for consciousness and when, if ever, these computations will exist in AI. [60] If this hypothesis is incorrect [61], then AI consciousness will remain at best a theoretical possibility until we move beyond current architectures.

For example, if more biology-like functions are required for consciousness, then AI consciousness may require novel hardware that can perform those functions. [62]

As we discuss below, our view is that computational functionalism is neither clearly correct nor clearly incorrect at this stage. We might lean one way or the other, but given the importance

and difficulty of consciousness as a research topic, we should leave room for doubt. That means

that AI consciousness assessments will need to be probabilistic rather than all or nothing at present

and, plausibly, for the foreseeable future. To use a simple example, if we estimate that there is a

30–50% chance that computational functionalism is correct and a 30–50% chance that an AI system

57Most saliently, what we learn from the human case is, most directly, what some of the sufficient conditions are for consciousness. It is difficult to know which elements we should take to be strictly necessary.

58Seth and Bayne (2022); Chalmers (2004)

59Computational functionalism is one kind of functionalism (Piccinini, 2018). For a survey of varieties

of functionalism, see Block (2009); Maley and Piccinini (2013). See Putnam (1967) for a classic statement

of computational functionalism. See Colombo and Piccinini (2023) and Rescorla (2020) for computational theories of mental phenomena more generally.

60Note that the theories of consciousness we consider, and the investigations of AI systems that we propose,

do not purport to solve the “hard problem” of consciousness, which concerns how physical processes relate to conscious experiences. Whatever mysteries there may be about this fundamental issue, virtually everyone agrees that physical processes (such as certain patterns of neural firing or certain computations) and conscious experiences are closely related in systematic ways. The theories and investigations at hand seek to find the neural processes and/or computations that are associated with consciousness.
61 Godfrey-Smith (2016, 2020); Cao (2022); Seth (2021, 2024)
62 Brunet and Halina (2020)

is conscious if so, then it follows that there is a 9–25% chance that this AI system is conscious. That

would be good to know when interacting with this AI system!

In this precautionary spirit, some of the authors of this report (Patrick Butlin, Robert Long, and

Jonathan Birch) released a paper in 2023 exploring the implications of several prominent scientific theories of consciousness [63] - viewed through the lens of computational functionalism — for AI

consciousness. The table below lists the theories and conditions we surveyed:

Recurrent processing theory

1.1 Input modules using algorithmic recurrence

1.2 Input modules generating organised, integrated perceptual representations

Global workspace theory

2.1 Multiple specialised systems capable of operating in parallel (modules)

2.2 Limited capacity workspace, entailing a bottleneck in information flow and a selective attention

mechanism

2.3 Global broadcast of information in the workspace to all modules

2.4 State-dependent attention, giving rise to the capacity to use the workspace to query modules in

1
2 succession to perform complex tasks

3
4 Computational higher-order theories

5
6 3.1 Generative, top-down or noisy perception modules

7
8 3.2 Metacognitive monitoring distinguishing reliable perceptual representations from noise

9
10 3.3 Agency guided by a general belief-formation and action selection system, and a strong
11 disposi

12 tion to update beliefs in accordance with the outputs of metacognitive monitoring

13
14 3.4 Sparse and smooth coding generating a ‘quality space’

15
16 Attention schema theory

17
18 4.1 A predictive model representing and enabling control over the current state of attention

19
20 Predictive processing

21
22 5.1 Input modules using predictive coding

23
24 Agency and embodiment

25
26 6.1 Minimal agency, that is, the capacity to learn from feedback and select outputs in such a
27 way as

28
29 to pursue goals, especially involving flexible responsiveness to competing goals

30
31 6.2 Embodiment, that is, the capacity to model output-input contingencies, including some
32 system

33 atic effects, and to use this model in perception or control

34
35 63Butlin et al. (2023)

36
37 These theories come in numerous versions. Weaker versions are directed only at
38 distinguishing

39
40 conscious from unconscious states in humans. For our purposes, what matters are stronger
41 versions

42
43 that aim to give sufficient conditions for consciousness across human and nonhuman
44 systems. The

1 crucial claim is that there is a realistic possibility that such a theory is correct, at least to the
2 extent

3
4 that the indicators associated with one or more theories jointly provide sufficient conditions
5 for

6
7 consciousness. Given this, and given that the indicators can be implemented in near-term AI
8 systems,

9
10 then there is a realistic possibility that near-term AI systems will be conscious.

11
12 It is by no means obvious that any of these theories are correct, especially in their stronger
13 versions. Our claim is not that they are correct, but only that there is a realistic possibility
14 that one

15
16 of them is correct. In fact, for our purposes, it would suffice if an extension of one of these
17 theories

18
19 is correct, or if some other computational theory is correct, as long as the relevant theory
20 provides

21
22 sufficient conditions that can be implemented in near-future AI.

23
24 After surveying the indicators and a variety of AI systems and methods, we found no clear
25 barriers to satisfying these indicators using current AI architectures and methods. And while
26 no

27
28 current systems seemed very likely to be conscious at the time, there do seem to be plausible
29 routes

30
31 towards conscious AI, according to these theories and assumptions.

32
33 Consider global workspace theory, which associates consciousness with a global workspace
34 —

35
36 roughly, a system that integrates information from mostly-independent, task-specific
37 information
38 processing modules, then broadcasts it back to them in a way that enables complex tasks like
39 plan

40 ning. We found that, as argued by Juliani et al. (2022), some AI architectures already
41 embody some,
42 though not all, aspects of a global workspace. [64] We also found that several technical
43 research programs seem poised to implement further aspects of a global workspace in the
44 near future. [65] And

indeed, researchers investigating consciousness in AI recently have subsequently built a system which aims to implement all of the global workspace indicators from Butlin et al. (2023). [66] We found similar trends for other theories as well. [67]

For the global workspace and other potential indicators of consciousness, progress might continue via direct efforts, in which researchers try to build conscious AI intentionally. Some researchers are motivated to build conscious AI because they view conscious AI as an end in itself.

“It would be monumentally cool,” as Graziano (2017, p. 7) puts it. Others are motivated to build

conscious AI because they believe that consciousness or related features could make AI systems safer or more capable. [68]

AI companies might also build conscious AI unintentionally. You might wonder why an AI system designed to, say, navigate a warehouse or manage a factory would be conscious. The answer

is that on some views of consciousness, the building blocks of consciousness can emerge as a side

effect of other cognitive capacities, such as perception, cognition, and robust agency. For example,

many theories of consciousness take metacognition — roughly, the ability to model your own cognition — as important for both consciousness and decision-making. [69] This kind of connection is also

64Similarly, Goldstein and Kirk-Giannini (forthcoming) have argued that language agents instantiate a global workspace in the relevant sense.

65VanRullen and Kanai (2021); Goyal et al. (2022); Goyal and Bengio (2022); Bao et al. (2020)

66Dossa et al. (2024)

67See Butlin et al. (2023), section 3.

68Bengio (2019); Goyal and Bengio (2022); Graziano (2017); Verschure (2016)

69Shea et al. (2014). For an overview of higher-order theories of consciousness, see Carruthers and Gennaro

(2020). As David Chalmers has noted (Chalmers (2018)), “In general, we should expect any intelligent system

1 to have an internal model of its own cognitive states.”

2
3 an element of, among others, global workspace theory, [70] higher-order theories, and the
4 attention

5
6 schema theory.

7
8 The link between consciousness and other cognitive capacities is unclear, and we certainly do
9
10 not want readers to naively equate them. But it may be that increased cognitive capacity
11 tends to

12
13 bring about consciousness, in both biological and digital systems. Thus, it may even be that
14 general
15 intelligence will itself be a plausible indicator for AI consciousness in the future. [71]

16
17 For these and other reasons, we agree with a recent open letter by consciousness scientists
18 and AI

19
20 researchers that “it is no longer in the realm of science fiction to imagine AI systems having
21 feelings
22 and even human-level consciousness.” [72] As one of the authors of this report, David
23 Chalmers, wrote

24
25 in 2023:

26
27 I think it wouldn’t be unreasonable to have a credence over 50 percent that we’ll

28
29 have sophisticated LLM+ systems (that is, LLM+ systems with behavior that

30
31 seems comparable to that of animals that we take to be conscious) with all of
32 these properties [73] within a decade. It also wouldn’t be unreasonable to have at

33
34 least a 50 percent credence that if we develop sophisticated systems with all of

35
36 these properties, they will be conscious. Those figures would leave us with a
37 credence of 25 percent or more [emphasis ours]. [74]

38 39 2.3 Robust agency in near-future AI

40
41 The robust agency-based case for expecting moral patienthood in near-term AI systems is
42 that there

43
44 is a realistic, non-negligible possibility that:

45
46 **1. Normative:** Robust agency suffices for moral patienthood, and

2. Descriptive: There are computational features — like certain forms of planning, reasoning, or self-awareness — that both:

- Suffice for robust agency, and
- Will exist in some near-future AI systems.

We will present arguments for each of these premises in turn.

2.3.1 Does robust agency suffice for moral patienthood?

The word “agency” is used in many different ways as well. In a broad sense, one might use “agent” to mean any entity that senses the environment and responds, [75] which would include thermostats,

70Shea and Frith (2019)

71Shevlin (2020). A related view is Danaher’s (2020) “ethical behaviorism,” which holds that we should

treat an entity as a moral patient if it is “performatively equivalent to other entities that have significant moral status.” Danaher stresses equivalence of behavior rather than equivalence of cognitive capacities.

72AMCS (2023)

73With “all of these properties,” Chalmers is referring to a number of proposed necessary conditions for consciousness, quite similar to those discussed here, that he argues will plausibly be satisfied by near-term AI systems: senses, embodiment, world-models, self-models, recurrent processing, global workspace, and unified agency.

74Chalmers (2023a). Chalmers also notes that these credences follow from mainstream views about consciousness; he adds that his own credences are higher, given that his own views about consciousness are more expansive.

75Russell et al. (2010, p. 34): “an agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through activators.”

or any entity that learns and pursues goals, which would include very simple RL agents that play

Tic-Tac-Toe. Some might take even these basic ways of being an agent to be sufficient for moral

1
2 patienthood and, to the extent that you accept this view, you likely already endorse moral
3 consider
4 ation for AI systems without needing to read further. While this view merits consideration
5 (and we

6
7 consider it further in upcoming work), we set it aside for now. We will be arguing that AI
8 systems
9 could be agents in a more demanding sense. [76]

10
11 Specifically, “robust agency” is the ability to pursue goals via some particular set of
12 cognitive

13
14 states and processes. Which ones? There are several “levels” of agency that extend beyond
15 the mere

16
17 ability to learn and pursue goals, and that could plausibly suffice for moral patienthood even
18 when
19 consciousness is absent. [77] For present purposes, we highlight three such levels:

20
21 **1. Intentional agency:** This is the capacity to set and pursue goals via beliefs, desires, and
22 intentions. Roughly, if you have mental states that represent what is, ought to be, and what
23 to do, and if these mental states work together in the right kind of way to convert perceptual
24 inputs to behavioral outputs, then you count as an intentional agent.

25
26
27
28
29 **2. Reflective agency:** This is intentional agency plus the ability to reflectively endorse your
30 own beliefs, desires, and intentions. Roughly, if you can form beliefs, desires, and in
31 tentions about your own beliefs, desires, and intentions, accepting or rejecting your own
32 attitudes and behaviors at a higher level, then you count as a reflective agent.

33
34
35
36 **3. Rational agency:** This is reflective agency plus the ability to rationally assess your own
37 beliefs, desires, and intentions. Roughly, if you can consider whether particular beliefs,
38 desires, intentions, actions are justified and adopt principles that you can treat as rules of
39 conduct, then you count as a rational agent. [78]

40
41
42
43 There might also be different ways of realizing each capacity, with different kinds and
44 degrees
45

of cognitive sophistication. For example, typical adult humans have the capacity for propositional thought, which means that our thoughts can have a structure that allows for truth values and logical relations. This capacity unlocks powerful forms of intentional agency, reflective agency, and rational agency, since it allows us to develop a wide range of novel beliefs, desires, and intentions and then use evidence and reason to assess their accuracy and coherence. In contrast, while nonhuman animals appear to lack this capacity, many animals at least have a limited capacity for symbolic thought, metacognition, and planning and problem solving. [79]

76Wooldridge and Jennings (1995); Schlosser (2019); Kenton et al. (2023)
 77For another taxonomy of agency for the AI context, see Dung (2024), who also notes the importance of agency for AI moral patienthood.
 78For any kind of agency, one key question will be whether that form of agency requires consciousness. For example, when assessing whether an AI system has beliefs and desires, one view would be that it is enough to have states that play the functional role of beliefs and desires (roughly, representing what is and what ought to be). On this view, beliefs and desires would not require consciousness. But another view would be that beliefs and desires might require that it feels like something for the AI system to perform those functions — roughly, it needs to have conscious feelings about how the world is and ought to be. On this view, beliefs and desires would require consciousness. For an overview of phenomenal intentionality, see Bourget (2019).
 79For example, many animals appear to have thoughts that take the form of maps and charts, with abstract information about social and environmental structures. Many animals also have perceptual affordances that represent what is, what ought to be, and what to do; a limited ability to represent their own mental states and the mental states of others; and a limited ability to make basic inferences based on disjunctions and negations.

1 These capacities allow for a limited form of all three levels of robust agency, with important
2 similarities with
3 and differences from the typical human form. For discussion, see Sebo (2017); Camp (2009);
4 Bermúdez (2009);
5 Gennaro (2009); DeGrazia (2009).
6

7 Why might these kinds of agency suffice for moral patienthood? First, intentional agents are
8 potentially welfare subjects. [80] It seems plausible that when you have desires, your life
9 goes better for

10
11 you when your desires are satisfied and worse for you when your desires are frustrated.
12 Moreover,

13
14 the satisfaction or frustration of desires can benefit or harm you whether or not you
15 consciously
16 experience them. [81] On some views, this is why we can be posthumously harmed, for
17 example. [82]

18
19 And while some people think that desire-satisfaction and desire-frustration matter only for
20 conscious

21
22 beings, others think that they matter for non-conscious beings as well, and so animals with
23 desires
24 deserve moral consideration whether or not they can consciously experience pain, for
25 instance. [83]

26
27 Reflective agency, particularly in its propositional form, then adds the ability to have desires
28
29 about our own desires, which is at the root of some conceptions of free will, the self, and
30 personal
31 identity. [84] Specifically, our desires become “ours” in a new sense when we endorse them
32 through

33
34 reflection. Reflective agency also allows for new kinds of morally significant interests and
35 relation
36 ships. When you can have mental states about other mental states, you can have at least a
37 limited

38
39 conception of how you and others think or feel. For example, reflective agents can have
40 preferences

41
42 about how they relate to each other, and all else being equal, their lives are better for them
43 when

44
45 those preferences are satisfied and worse for them otherwise.
46

1 Rational agency, particularly in its propositional form, then adds the ability to create social
2
3 contracts with other rational agents (assuming that they can communicate as well), which are
4 at the
5 root of some conceptions of moral, legal, and political rights and responsibilities. [85]
6 Rational agency

7
8 also allows for decisions based on judgments about reasons and principles, and this ability
9 not only

10
11 allows for new kinds of interests, but also — on some views — commands a kind of respect
12 that

13
14 extends beyond compassion. Indeed, this idea of respect for rational agents is at the root of
15 the

16
17 Kantian ethical theory, which rests alongside the utilitarian idea of beneficence for sentient
18 beings
19 as one of the two most influential ethical theories in the modern era. [86]
20

21 In what follows, we discuss the route towards robust agency in general rather than the routes
22
23 towards intentional, reflective, and rational agency in particular. We distinguish these levels
24 of robust

25
26 agency here to emphasize that there can be different kinds of robust agency with different
27 kinds of

28
29 moral significance, both within and across these levels. That means that when we search for
30 robust

31
32 agency in nonhumans, including animals and AI systems, it would be a mistake to anchor too
33 much

34
35 on human agency. But having now made this point, we focus on showing that AI
36 development is

37
38 80This kind of view reflects one of the main philosophical theories of welfare, known as the
39 desiresatisfaction view. This view holds that your life goes better or worse for you to the
40 extent that your desires
41 are satisfied or frustrated, independently of whether you consciously experience the
42 satisfaction or frustration
43 of your desires. See, for instance, Heathwood (2015).

44 81Dorsey (2013)

45 82See, for example, Nozick (1974); Rachels (1986); Kagan (1994); Ruddick (2005).

83See Neely (2014); Kagan (2019); Kammerer (2022); Delon (2024); Goldstein and Kirk-Giannini

(forthcoming), who consider and/or defend versions of this view.

84Frankfurt (1971)

85Traditionally, philosophers have focused more on the significance that these capacities might have for

moral agency. This is a different concept from moral patienthood: In general, you are a moral agent if you

can have duties to others, and you are a moral patient if others can have duties to you. And many philosophers

believe that a particular kind of agency — namely, rationality — is important for moral agency because you

need to be able to reason about your actions in order to be morally accountable or responsible for them. When

a non-rational entity (say, a hurricane) kills someone, we can call it bad. But when a rational agent knowingly

and willingly kills someone, we also call it wrong.

86However, please note that while Immanuel Kant (1785) accepted rationality as the basis for moral patienthood, some contemporary Kantians, such as Korsgaard (2018), now accept sentience as the basis for moral patienthood instead.

currently on a path to create many computations that we associate with all of these levels of robust

agency.

2.3.2 Will some AI systems be robustly agentic in the near future?

There are plausible routes by which we might soon have AI systems with robust agency. What it

takes to be an intentional, reflective, and/or rational agent is not clear. Many cognitive capacities

are plausibly associated with each level — capacities that are the aims of well-resourced research

programs that are making significant progress. For example, intentional agency is related to capacities

for planning, memory, and learning. Reflective agency is related to introspection and situational

awareness. And rational agency is related to all of these capacities, along with abstract reasoning

1 about principles and strategies.

2
3 The capacities described in this section might or might not be necessary or sufficient for
4 these

5
6 levels of robust agency. But to the extent that these capacities are present, the probability of
7 robust

8
9 agency will increase. And the development of AI systems with such capacities aligns with
10 the

11
12 aims of significant research and development efforts: Both major tech companies and
13 startups are
14 investing heavily in creating more agentic AI systems, with significant progress already
15 evident. [87] In

16
17 what follows, we discuss recent work in reinforcement learning, language agents, and other
18 research
19 programs as case studies. [88]

20
21 First, reinforcement learning (RL) is the subfield of AI most concerned with building agents
22 as a fundamental goal, as Sutton and Barto put it in their canonical RL textbook:

23
24] explicitly considers the whole problem of a goal-directed agent interacting
25 with an uncertain environment. [89]

26
27
28 In RL, the aim is to write algorithms that allow agents to learn, reason, and act in pursuit of
29
30 specified goals in complex environments. RL construes goal-pursuit as maximizing reward
31 through

32
33 interaction with the environment, and some RL researchers argue that this process allows
34 agents to
35 acquire the whole suite of capacities observed in intelligent systems. [90]

36
37 RL is a useful starting point for examining robust agency in AI, not only because of its promi
38 nence and recent successes, but also because of its centrality of reinforcement learning for
39 human
40 and nonhuman agency. [91] Of course, in humans and many other animals, a rich
41 understanding of

42
43 social and environmental context and the expressive power of language, among other
44 capabilities,
45

1 make substantial contributions to our capacities for robust agency as well. But deep RL has
 2 made
 3
 4 it possible for AI agents to be virtually embodied and situated in environments comparable to
 5 those
 6 inhabited by animals, [92] and so it may be a compelling foundation for projects to emulate
 7 natural
 8
 9 agency.
 10
 11 Deep RL has achieved significant successes in game-playing in the last decade. These
 12 successes
 13 include superhuman performance in Go, chess, shogi, and a variety of Atari games (i.e.
 14 MuZero), [93]
 15
 16 87Toner et al. (2024)
 17 88We focus on the research programs that seem in recent years to be making the most
 18 strides, though doubtless there have been and likely will be other paradigms that make
 19 progress.
 20 89Sutton and Barto (2015, p. 3)
 21 90Silver et al. (2021)
 22 91Dolan and Dayan (2013)
 23 92Shanahan et al. (2020); Abramson et al. (2021)
 24 93Schrittwieser et al. (2020)
 25
 26 as well as in games that involve controlling an avatar in a relatively rich and dynamic
 27 environment,
 28 such as Gran Turismo (GT Sophy) [94] and Starcraft II (AlphaStar). [95] Many of these
 29 agents, such as
 30
 31 MuZero, can learn models of the game environments (including opponent behavior) and use
 32 them to
 33
 34 make the predictions in order to plan. Some game-playing agents can also exploit a form of
 35 episodic
 36 memory, called experience replay, in order to increase sample efficiency. [96]
 37
 38 While many agents have relatively narrow capabilities, DeepMind’s Adaptive Agent
 39 demonstrates the ability to rapidly adapt to new tasks in a 3D virtual environment. [97] AdA
 40 is trained on
 41
 42 a varied curriculum of tasks, inducing meta-learning of an algorithm for few-shot learning of
 43 new
 44
 45 tasks — that is, for learning how to make reliable predictions and decisions based on a small
 46 number

of examples. The architecture includes a Transformer-based memory module encoding recent observations, allowing the system to identify dependencies between actions and subsequent events. As a

result, AdA has a notably effective, flexible way of acquiring and using a grasp of the environment's

dynamics.

RL research targets robust agency partly through research on abstraction and hierarchical planning. In complex environments that require intricate sequences of movements, it can be vital to use

representations that abstract away from low-level details. For example, Director can learn to break down tasks with sparse rewards into subgoals. [98] RL research also targets robust agency through

research on multiplayer strategy games such as Diplomacy, which involves forming alliances. For example, Meta's Cicero [99] uses an adapted language model to achieve comparable performance to

high-level human players, planning in ways that predict human behavior and changing plans through

communication with these other agents.

Researchers are now pursuing several promising strategies involving language agents as well. [100]

Language agents leverage the powerful natural language processing and generation abilities of

LLMs for greater capability and flexibility, by embedding LLMs within larger architectures that support functions like memory, planning, reasoning, and action selection. [101] While existing systems

struggle with reliability, the properties of language agents and their initial success suggest that they have the potential to overcome traditional barriers towards more agentic systems [102] . Indeed, several

notable examples of language agents have emerged in recent years, demonstrating that this strategy can lead to more robust and generalized agency across diverse domains: [103]

- ReAct (Yao et al., 2023) alternates between generating thoughts/plans and taking actions

in interactive environments. It can break down complex tasks, gather information dynam

94Wurman et al. (2022)

95Vinyals et al. (2019)

96Mnih et al. (2015)

97Team et al. (2023)

98Hafner et al. (2022)

99 (FAIR)

100Mialon et al. (2023); Wang et al. (2024); Sumers et al. (2024); Guo et al. (2024)

101Depending on one’s definition, even LLMs that are used as (or in) chatbots, like

ChatGPT, may also qualify

as “language agents” (see Butlin (n.d.) for discussion). Goldstein and Levinstein (2024)

argue that such LLMs

also plausibly have beliefs and desires, noting the importance of this claim for issues of moral patienthood.

102LLMs in their own right are also making progress towards agency, and in particular when combined

with RL methods. Recent work has explored using language models as the starting point for RL training,

as in reinforcement learning from human feedback (RLHF), LM-based proof-writing agents, and LM-based

coding agents (Ruan et al. (2024)). Indeed recent benchmarking work suggests progress on language modeling

capabilities naturally results in performance gains on agentic tasks. See Ziegler et al. (2020);

DeepMind (2024);

Gehring et al. (2024)

103See Butlin (n.d.), who discusses several of these systems in light of various conceptions of agency.

ically, and adjust its approach based on intermediate results. ReAct has shown strong

performance on language-based tasks like question answering and web navigation.

- Generative Agents (Park et al., 2023) simulates interactive AI characters. The agents have

persistent identities, relationships, and goals, with an LLM generating plans and actions

1 based on their memories, observations, and reflections. As a result, they exhibit long-term
2
3 coherence with evolving goals and emergent social behaviors.
4

5 – Voyager (Wang et al., 2023) uses an LLM to control an embodied agent in Minecraft,
6 iter
7 atively setting its own goals, devising plans, and writing code to accomplish increasingly
8
9 complex tasks. By maintaining a skill library and reflecting on past experiences, Voyager
10
11 can bootstrap its way to mastering the game's tech tree and creatively solving novel chal
12 lenges.
13

14 – SayCan (Ahn et al., 2022) grounds language in robotic control, using an LLM to
15 generate
16
17 high-level plans that are then mapped to concrete robot skills. This allows the system to
18
19 flexibly respond to natural language commands by reasoning about affordances and break
20 ing tasks into actionable steps.
21

22 These language agents, and others, exhibit several key properties that make them more
23 robustly
24

25 agentic compared to many traditional AI systems (though progress has certainly been made
26 outside
27

28 the language agent paradigm as well):
29

30 – Flexible goal-setting and planning: Rather than being constrained to predefined reward
31
32 functions, language agents can understand open-ended objectives, generate their own sub
33 goals, and devise multi-step plans to achieve them.
34

35 – Adaptive reasoning: By leveraging LLMs' broad knowledge and reasoning capabilities,
36 lan
37 guage agents can navigate novel contexts, drawing from relevant insights in other contexts
38
39 to inform their decisions.
40

41 – Memory integration: Many language agents incorporate episodic and semantic
42 memory
43
44 systems, allowing them to learn from experience, maintain consistent behaviors, and apply
45 past knowledge to new contexts. [104]
46

1 – Metacognition: Agents like Voyager and Generative Agents can reflect on their own
2
3 thoughts and experiences, enabling higher-order reasoning and self-improvement.
4

5 – Open-ended interaction: These systems can often engage in natural language dialogue,
6
7 explain their reasoning, and incorporate new information or instructions on the fly.
8

9 For these and other, similar reasons, Goldstein and Kirk-Giannini (forthcoming) argue that
10 philo
11 sophical consideration of language agents suggests that “the technology already exists to
12 create AI

13
14 systems with wellbeing.” Our conclusion at this stage is somewhat more tentative: Rather
15 than

16
17 assert or deny that the technology already exists to create such systems, we merely assert that
18 the
19 technology already exists to create key properties of such systems. [105] We also note that,
20 while many

21
22 104cf. also, among others, Rubin et al. (2022); Shinn et al. (2023).

23 105Language agents and other generative agents are clearly quite different from biological
24 agents. For example, their ‘observations’ are all in language, their ‘intelligence’ is mostly
25 embedded within language models,
26 and many of them do not learn via reward and punishment. It is hard to know exactly what to
27 say about them,
28 and we are not claiming that they have robust agency. Instead, we are claiming that they
29 constitute a significant
30 step towards robust agency in AI. While language agents and other generative agents are still
31 limited in many
32 ways, their ability to flexibly pursue goals, reason about abstract concepts, and adapt to novel
33 situations is a
34 striking indicator of what else may soon be possible.

35
36 “language agents” may turn out to be impressive demos that do not scale, frontier language
37 models
38 are also being made more agentic by the day [106] . The question is whether and to what
39 extent progress

40
41 will continue from here. We believe that the current state of the field, combined with the
42 clear in
43 centives that developers have for continuing to build towards robust agency, strongly suggest
44 that

45
46 progress will continue.

Specifically, we expect that future routes have the potential to produce further properties associated with robust agency. Consider an AI that combines the learning capabilities of RL agents with

the reflection and world modeling capabilities of LLMs. In a recent interview, Demis Hassabis discusses one possible example of such a system: combining LLMs with the Monte Carlo Tree Search (MCTS) used by, say, AlphaGo. [107] In such a system, the LLM provides a rich, flexible "belief"

system about the world. The LLM could be used to analyze the system's decision-making process,

approaching a form of meta-cognition. This system might also be able to provide explanations or justifications for its decisions, approaching a form of rational deliberation. [108]

Applying RL to augment run-time performance of LLMs is a promising route as well. Consider

OpenAI's o1 system. Described as a "Large Reasoning Model" (LRM), o1 incorporates extensive

chain-of-thought reasoning, which is speculated to be trained using RL. This approach, combined

with increased compute usage at inference time, has led to markedly improved performance on

planning and reasoning tasks, nearly saturating some benchmarks that previously challenged LLMs.

o1 also exhibits concerning behaviors related to instrumental convergence, deceptive alignment, and

reward hacking. These traits, while currently limited in scope and impact, highlight the potential for

LLMs to be used to make systems that are more agentic.

The long-term ambitions of AI research have consistently aimed towards creating systems that

exhibit key characteristics of robust agency. These goals include developing AI systems that
 can operate on extended time horizons, maintain coherent objectives, engage in self-reflection, and
 revise
 their own goals and methods. For large swaths of the field, the ultimate aim has been and
 remains to
 create "human-level" AI capable of general problem-solving across diverse domains. [109]
 This vision
 implicitly requires many of the capacities we associate with robust agency: intentional goal-
 setting,
 reflective self-assessment, and rational decision-making in complex, dynamic environments.
 These aims are also central to current investment and effort. For example, major tech
 companies
 like Microsoft and Google have announced plans for AI tools with “more autonomy and less
 human
 intervention” and agents that can autonomously carry out complex multi-step tasks. [110]
 Simultane-
 ously, startups such as Adept, MultiOn, and Lindy have raised hundreds of millions of
 dollars to
 develop flexible AI agents. This excitement could be overstated and/or misplaced, of course
 — and
 recent work has already seen periods of hype and disillusionment, as with systems like
 AutoGPT —
 but it is evidence that a lot of effort will be put into building more agentic AI systems in the
 coming
 years.

106Anthropic (2024)
 107Hassabis (2022)
 108This is just one kind of potential capability enhancement from synergies between LLMs
 and RL;
 Pternea et al. (2024) surveys a variety.
 109See, among many others, McCarthy et al. (1955); Moravec (1995), Russell et al. (2010,
 ch. 27),
 Morris et al. (2024).
 110Microsoft touts systems “that can now act as independent agents — ones that can be
 triggered by events

1
2 • not just conversation — and can automate and orchestrate complex, long-running business
3 processes with
4 more autonomy and less human intervention.” See Aftab (2024).
5

6 While today's dominant AI paradigm, centered around LLMs, often appears less explicitly
7 agen
8 tic, there are clear signs of a shift towards more agent-like systems like the kinds discussed
9 here.

10 Consider the goal of building advanced AI assistants that can undertake a variety of complex
11 tasks. [111]
12

13 Such a system would need to be able to set and pursue long-term goals based on high-level
14 reflections about facts and values. [112] Of course, there is no way of knowing for certain
15 whether current
16

17 efforts to build such a system will succeed. But there is a strong incentive to maintain these
18 efforts
19

20 and a tenable path toward success. Relevant actors thus have reason to consider not only the
21 benefits
22

23 but also the risks and harms that might follow from success.
24

25 2.4 Decision-making under uncertainty 26

27 These reflections identify two routes towards near-future AI moral patienthood. Again, the
28
29 consciousness-based case holds that there is a realistic possibility that (1) consciousness
30 suffices
31

32 for moral patienthood, (2a) some class of computational features suffice for consciousness,
33 and
34

35 (2b) some AI systems will have these features in the near future. The robust agency-based
36 case is
37

38 structurally identical, but with robust agency instead of consciousness. How should we assess
39 these
40

41 arguments? That, of course, depends on how confident we are in their premises. It also
42 depends on
43

44 how we make decisions in cases involving uncertainty about key normative and descriptive
45 issues.
46

1 This section briefly discusses these issues.

2
3 The premises of these arguments invoke difficult questions, and in future work we will
4 present

5
6 a research agenda that discusses these questions in more detail. But at a high level, it would
7 be a

8
9 mistake to reject any of these premises out of hand. These arguments address foundational
10 issues

11
12 in philosophy, science, and technology; issues involving what it takes to matter, what it takes
13 to

14
15 think and feel, and what the future of AI holds. We may well favor some views about these
16 issues

17
18 over others. But given how difficult and contested these issues are, we should embrace
19 caution and

20
21 humility about our current views, aspiring to learn more and preparing for the possibility that
22 our

23
24 views will change over time.

25
26 This kind of caution and humility is enough to motivate the recommendations that we make
27 in

28
29 this report. We are not arguing that near-future AI systems will, in fact, be moral patients, nor
30 are we

31
32 making recommendations that depend on that conclusion; that would require assessing these
33 issues

34
35 with more precision and reliability than we think is possible at present. We are instead
36 arguing that

37
38 near-future AI systems have a realistic chance of being moral patients given the information
39 and

40
41 arguments currently available, and we are making recommendations that depend on that
42 conclusion

- 43
44 • recommendations that focus on aspiring to learn more while preparing for the possible
45 emergence
46

of AI moral patienthood as a precautionary measure.

To see why we think that this kind of caution and humility is warranted at this stage, we briefly

consider three key uncertainties one might have about the prospect of near-future AI moral patienthood: first, about the bases of welfare and moral patienthood; second, about the bases of consciousness and robust agency; and third, about the path and pace of near-future AI progress. We then

consider how to make important decisions when confronted with uncertainty about multiple key issues at the same time. As we discuss below, substantial uncertainty about the nature and intrinsic

value of AI systems does not mean that we should postpone taking AI welfare seriously; on the

contrary, it means that we should take AI welfare seriously now.

¹¹¹See Gabriel et al. (2024); Dong et al. (2023)

¹¹²LeCun (2024)

2.4.1 What if these capacities are insufficient for moral patienthood?

We have focused on consciousness and robust agency because they are two of the most prominently

defended bases of welfare and moral patienthood. But this list of potential bases of welfare and

moral patienthood is far from comprehensive, and there is ongoing disagreement and uncertainty

about whether these capacities — individually or jointly — are necessary or sufficient.

The consciousness-based route holds that consciousness (either valenced or non-valenced) suffices for moral patienthood. But some views of moral patienthood are more demanding. For example,

on some views, moral patienthood requires reciprocity, which requires rationality; I can have

1 duties to you only if you can have duties to me, and you can have duties to me only if you
2 can rationally assess your actions. [113] However, this view implies that a wide range of
3 vulnerable beings lack

4
5 moral patienthood, including not only nonhuman animals but also infants and other non-
6 rational humans. [114] A more common view is that you need rationality to have particular
7 rights, like the right
8 to drink or smoke, but not to merit moral consideration in general. [115]

9
10 Meanwhile, the agency-based route relies on the premise that agency (either rational or non
11 rational) suffices for moral patienthood, even without consciousness. This view holds that
12 con
13 sciousness is not the only way to have a morally significant subjective perspective; the
14 possession
15 of beliefs, desires, intentions, and other such states is another. [116] But this view is far from
16 secure,

17
18 since many philosophers still hold that moral patienthood requires consciousness, either
19 valenced or

20
21 non-valenced. So our agency-based route relies on a more controversial basis of moral
22 patienthood

23
24 than the consciousness-based route does. Still, it would be rash to dismiss this view entirely
25 as we
26 create increasingly complex agents. [117]

27
28 With that said, we should also keep in mind that some AI systems could be both conscious
29 and

30
31 agentic in relevant respects. And the idea that these capacities jointly suffice for moral
32 patienthood

33
34 is both very plausible and widely accepted. That is, if AI systems could experience happiness
35
36 and suffering and set and pursue their own goals based on their own beliefs and desires, then
37 they

38
39 would very plausibly merit moral consideration (though their interests and rights could still
40 be quite

41
42 different from ours). Granted, some restrictive views about moral patienthood would deny
43 this claim,

44
45 such as views that require membership in the species *Homo sapiens*. But it would be
46 especially rash

to dismiss nonhuman moral patienthood on such grounds.

2.4.2 What if these features are insufficient for these capacities?

We have also focused on certain computational features that might suffice for consciousness and/or

robust agency. But some theories of consciousness and robust agency give an essential role to biology or to other features that current AI systems lack. [118] (Here, we focus on these sorts of objections regarding consciousness). [119]

113See Carruthers (2011); Wissenburg (2014), among others. cf. Korsgaard (2018, ch. 7).

114Andrews et al. (2018, pp. 17–47)

115Andrews et al. (2018, pp. 85–114), Korsgaard (2018)

116Levy (2024)

117cf. Goldstein and Kirk-Giannini (2024), who discuss risk in the context of their agency-centric case for AI wellbeing.

118One cluster of views we do not focus on here are theological views that assign importance to a non-physical soul. See Turing (1950) for a canonical discussion, recently elaborated upon by Cutter (forthcoming).

119For a recent defense of the necessity of biology for agency, see Jaeger et al. (2024).

There are a variety of views that would rule out AI consciousness on existing hardware. Some

views hold that consciousness requires biology in principle — if a system is nonbiological, then it is

nonconscious, no matter what computations it performs. [120] Other views hold that consciousness re

quires computational features that, at least at present, require biology in practice — such as specific

kinds of oscillations that require specific kinds of chemical and electrical signals. [121] And still other

views hold that consciousness requires computational features that, while perhaps possible in nonbiological systems, are still not present in mainstream AI hardware — such as analog computations [122]

or integration of computations across space and time. [123]

1 As in past work, we do not take computational functionalism to be clearly true, nor do we
2 take
3
4 any of these alternatives to have been refuted. Our position is that computational
5 functionalism is
6
7 plausible and well-supported, and so it would be a mistake to dismiss near-future AI welfare
8 and
9
10 moral patienthood solely on the basis of high-level arguments against this assumption. If, in
11 the near
12
13 future, we built AI systems on existing hardware that possessed all of the computational
14 features
15
16 described in sections 2.2.2 and 2.3.2, then we might not be warranted in being very confident
17 that
18
19 consciousness and robust agency are present. But we would also not be warranted in being
20 very
21
22 confident that these capacities are absent.
23
24 Expert surveys support the value of keeping an open mind about the basis of these capacities
25
26 at this stage. For example, in a survey of members of the Association for the Scientific Study
27
28 of Consciousness, only 3% responded “no” to the question: “At present or in the future,
29 could
30
31 machines (e.g., robots) have consciousness,” and over two thirds of respondents answered
32 “yes” or
33 “probably yes.” [124] And in a 2020 survey of professional philosophers, around 39%
34 responded that
35 they accept or lean toward the view that future AI systems will be conscious. [125] If our
36 confidence in
37
38 the possibility of AI consciousness is anywhere in the ballpark of these percentages — as, we
39 think,
40 it should be — then that is more than enough for our purposes here. [126]
41
42 2.4.3 What if these routes encounter a roadblock?
43
44 AI has progressed significantly in the last decade, and our arguments in this report consider
45 the
46

possibility of further progress in the decade to come. But of course, the path and pace of further AI

For example, according to a physicalist biological substrate theory, consciousness is simply identical to states or processes of biological, carbon-based neurons (see Hill (1991, pp. 10–12), Block (2009); McLaughlin (2012), Block (2023, pp. 445–446)). These views entail that no silicon-based system can be conscious as a matter of principle.

Peter Godfrey-Smith, who has argued for such a view, is “skeptical about the existence of non-animal” consciousness at present, including AI consciousness (Godfrey-Smith, 2020), though he also notes that his view “would not suggest a barrier to artificial consciousness per se, but a need for new architectures if such systems were to be built” (Godfrey-Smith, 2024). Other theorists express skepticism about AI consciousness on current hardware for similar reasons (Seth, 2021; Shiller, 2024). Brunet and Halina (2020) argue that the hardwares of (most) current AI systems do not satisfy the functional criteria outlined by Godfrey-Smith (2016), but that some existing and future hardwares might.

Arvan and Maley (2022)

Shiller (2024). Proponents of integrated information theory (IIT) make a similar argument, though for different reasons (Koch, 2019).

Francken et al. (2022)

Bourget and Chalmers (2023)

To be clear, we are definitely not arguing that one must always defer to expert views — especially in cases like this, when the relevant field is not particularly mature. But we do take these results to be part of a case against premature dismissal of the possibility of AI consciousness.

progress depend on a wide range of social, political, economic, and technological factors, and we have substantial disagreement and uncertainty about these factors as well. [127]

On the one hand, AI progress could slow down, even stall. AI companies have made significant

progress in recent years due in large part to the scaling of large models. However, there could be

1 diminishing returns with further scaling, particularly for tasks that require long time horizons.
2 There

3
4 could also be a “data wall” or other technical roadblocks, without any algorithmic or
5 architectural

6
7 breakthroughs that allow us to circumvent them. And there could be economic resistance to
8 further

9
10 scaling due to how expensive and resource-intensive this process is becoming (especially if
11 capabil
12 ity gains are insufficient to motivate continued investment), as well as political resistance or
13 other

14
15 societal disruptions that slow or stall progress.

16
17 On the other hand, AI progress could also continue at its current pace, or even speed up sig
18 nificantly. There could be continued or increasing returns from further scaling. There could
19 also

20
21 be algorithmic and architectural breakthroughs that reveal new pathways towards progress;
22 AI sys
23 tems have already started to contribute to AI research, and further capability gains in coding
24 and
25 reasoning could accelerate this feedback loop. [128] And there could be increased economic
26 support [129]

27
28 for further development due to the anticipated financial benefits of creating advanced AI,
29 and/or

30
31 increased political support for further development, potentially exacerbated by international
32 AI race

33
34 dynamics.

35
36 At present, nobody knows for sure whether AI progress will slow down, continue at its
37 current

38
39 pace, or speed up. However, we can make two observations here. First, when in doubt about
40
41 how a powerful technology will develop, we should plan for all realistic possibilities,
42 including

43
44 the possibility of significant progress. Second, even if AI progress did slow down from here,
45 our
46

recommendations would still stand. Existing AI systems already possess indicators of consciousness and robust agency, and developers could further integrate and amplify these capabilities even barring significant further progress. For these reasons, the fact that progress could slow or stall is compatible with the need for reasonable precautionary measures today.

2.4.4 What if the probability of AI welfare and moral patienthood is low?

These reflections raise the question how to make decisions about AI welfare under substantial uncertainty. We cannot be certain at this stage that the premises of these arguments are true or false;

instead, we can have only higher or lower degrees of confidence. What if these estimates together

imply that the probability of AI welfare and moral patienthood is low? As a toy example, suppose

there is only a ~25% chance that sentience suffices for moral patienthood, a ~25% chance that certain computations suffice for sentience, and a ~25% chance that some AI systems will be capable

of these computations in the near future. Assuming these chances are independent, it would follow

that there is only a ~2% chance of near-future AI welfare and moral patienthood via the sentience

route!

¹²⁷See Park et al. (2023) on the key arguments for why AI capabilities from scaling will, or will not, plateau soon.

¹²⁸See Woodside (2023)

¹²⁹Recent economic analysis from Epoch AI indicates that at the current exponential rate of expenditure increase on AI training, there will be no bottleneck to scaling up the inputs to AI training through 2030.

Sevilla et al. (2024)

1
2 Here we make two main observations. First, we expect that reasonable assessments of these
3 arguments will yield higher estimates. [130] In our view, for example, it would be reasonable
4 to hold

5
6 that there is a ~90% chance that sentience suffices for welfare and moral patienthood, a
7 ~50% chance

8
9 that certain computations suffice for sentience, and a ~50% chance that some AI systems will
10 be

11
12 capable of these computations in the near future. Assuming independence, it would follow
13 that there

14
15 is a ~22.5% chance of AI near-future welfare and moral patienthood via the sentience route
16 alone.

17
18 And if we then consider other possible sufficient conditions for welfare and moral
19 patienthood, such

20
21 as consciousness or various kinds of robust agency, then that might strengthen the case
22 further.

23
24 Second, even if the chance of near-future AI welfare and moral patienthood were as low as
25 2%,
26 that would still constitute a non-negligible risk. [131] Yes, if the chance was only, say, one in
27 a hundred

28
29 million, then we could debate whether that risk is low enough to ignore. But when the chance
30 of

31
32 near-future AI welfare and moral patienthood is at least one in a hundred (and, again, we
33 expect it to

34
35 be higher), such a debate is hard to justify. In other policy domains, we recognize that if the
36 chance

37
38 of a potentially large-scale harm rises to this level, we ought to assess this risk further and
39 prepare

40
41 a reasonable policy response. In short, this is not a “there may be an alien invasion soon”
42 kind of

43
44 chance. This is a “there may be another pandemic soon” kind of chance.

45
46 Of course, one could attempt to argue that the chance of near-future AI moral patienthood is

1
2 lower still — low enough to be at least arguably negligible. But while this claim might seem
3
4 plausible when we think about the issue in abstract terms, leaning on our intuitions about
5 which
6
7 kinds of beings can matter, it becomes less plausible when we think about the issue in more
8 concrete
9
10 terms, taking into account the current state of uncertainty in relevant subfields of philosophy,
11 science,
12
13 and technology. In short, our current epistemic situation calls for caution and humility. We
14 might
15
16 lean one way or the other, but we should keep an open mind and take reasonable steps to
17 prepare for
18
19 the possibility that our current views are mistaken.
20

21 3 Recommendations for AI companies

22 3.1 Introduction

23
24
25 We now present our recommendations for leading AI companies about how to respond to the
26 realistic,
27
28 non-negligible chance that some near-future AI systems will be welfare subjects and moral
29 patients.
30

31 We focus on first steps that AI companies can take within the next year, and they fall into
32 three

33
34 general categories:

- 35
36 – Acknowledge. Acknowledge that AI welfare is an important and difficult issue, and
37 that

38
39 there is a realistic, non-negligible chance that some AI systems will be welfare subjects and
40
41 moral patients in the near future. That means taking AI welfare seriously in any relevant
42
43 internal or external statements you might make. It means ensuring that language model
44
45 outputs take the issue seriously as well.
46

1 – Assess. Develop a framework for estimating the probability that particular AI systems
2
3 are welfare subjects and moral patients, and that particular policies are good or bad for
4 them. We [132] have templates that we can use as sources of inspiration, including the
5 “marker

6
7 130For further discussion, see Sebo and Long (2023). For more general discussion of
8 decision-making under
9 uncertainty, see Monton (2019).

10 131Sebo and Long (2023); Sebo (2025)

11 132As noted in footnote 12, here we use the collective “we” to refer to the constellation of
12 actors that have a
13 role to play in this work, including researchers, companies, and governments.

14
15 method” that we use to make estimates about nonhuman animals. We can consider these
16
17 templates when developing a probabilistic, pluralistic method for assessing AI systems.

18
19 – Prepare. Develop policies and procedures that will allow AI companies to treat
20 potentially

21
22 morally significant AI systems with an appropriate level of moral concern. We have many
23
24 templates to consider, including AI safety frameworks, research ethics frameworks, and
25
26 forums for expert and public input in policy decisions. These frameworks can be sources
27
28 of inspiration — and, in some cases, of cautionary tales.

29
30 We do not recommend relatively high-cost actions here, such as committing to halt
31 development

32
33 and deployment when red lines are crossed. Instead, we focus here on low-cost actions that
34 will

35
36 empower AI companies to make decisions about relatively high-cost actions thoughtfully in
37 the

38
39 future. In taking these steps, AI companies can work with experts, the public, and other
40 stakeholders
41 to identify further actions that can be taken in the near future. [133]

42
43 Before we proceed, we should emphasize that while our recommendations here focus on AI
44
45 welfare, AI safety remains a key priority as well. In upcoming work we discuss the
46 relationship

1
2 between AI safety and AI welfare in more detail, with reflections about how to pursue these
3 goals

4
5 simultaneously. For now, we focus on low-cost procedural recommendations for AI welfare
6 that are

7
8 compatible with, if not beneficial for, similar work in AI safety.

9
10 We should also emphasize that while we focus on LLMs for the sake of simplicity and
11 specificity

12
13 here, most of our recommendations apply to other kinds of AI systems too. Indeed, acting on
14 these

15
16 recommendations might be all the more pressing for other kinds of AI systems, since we
17 might be

18
19 more at risk of overlooking the potential moral significance of an AI system when that AI
20 system is

21
22 not designed to look or act like a human.

23 24 3.2 Acknowledge

25
26 As a starting point, AI companies have a responsibility to acknowledge that AI welfare is an
27 im
28 portant and difficult issue, and that there is a realistic, non-negligible chance that some AI
29 systems

30
31 will be welfare subjects and moral patients in the near future. As noted above, that means
32 taking

33
34 this issue seriously in any relevant internal or external statements they might make. It also
35 means

36
37 ensuring that LLMs take the issue seriously in any relevant statements they might make. In
38 short, if

39
40 and when leaders in this space discuss AI welfare and moral patienthood, they should make it
41 clear

42
43 that this is not merely a topic for science fiction, or a risk for the far future. This is a risk for
44 the near

45
46 future, and we should start taking steps to consider and mitigate it now.

Communicating about this topic requires careful calibration. There are significant risks associated with overattributing and underattributing welfare and moral patienthood to AI systems. So it

would be a mistake for AI companies to respond to overattribution risks by simply denying that AI

systems are welfare subjects and moral patients, and it would also be a mistake for them to respond

to underattribution risks by simply asserting that AI systems are welfare subjects and moral patients.

Instead, AI companies will need to strike a careful balance, by expressing uncertainty about this

topic while reassuring the public that we have tools that we can use to consider and mitigate risk in

such cases.

As discussed in section 1.1 and in section 3.4, we present these recommendations as the minimum first

steps that AI companies should take regarding AI welfare. However, we believe that other actors have a responsibility to take this issue seriously as well, and that AI companies — along with other actors — will have a responsibility to take further steps in the future.

In this section we make high level recommendations for how AI companies can strike this balance in their own communications, as well as for how they can train their LLMs to do the same.

We focus less on when companies should communicate about the issue — for example, whether to

communicate before or after taking steps toward developing internal policies and procedures — and

more on how companies should communicate about the issue. However, we do briefly note that companies will likely need to communicate about the issue sooner rather than later. And their models

1 are already communicating with users about it every day. So time is of the essence for taking
2 these
3
4 steps.

5 6 3.2.1 Recommendations for companies 7

8 Initial statements about AI welfare and moral patienthood can focus on making two basic
9 points:

10
11 – AI welfare and moral patienthood is both important and difficult. Humans have a
12 tendency
13
14 to mistakenly see subjects as objects and a tendency to mistakenly see objects as subjects,
15
16 and both of these mistakes can be harmful. Avoiding these mistakes requires engaging with
17
18 challenging problems in philosophy, science, and technology, about which there is substan-
19 tial disagreement and uncertainty among experts and non-experts alike. We thus need to
20
21 assess these issues carefully and thoughtfully, rather than simply dismissing them because
22
23 of their novelty, or relying on our own current (possibly biased) intuitions or judgments
24
25 about them.

26
27 – While the probability of AI welfare and moral patienthood might be low at present, it
28
29 will increase over time. Given current evidence, there is at least a realistic possibility that
30
31 (a) sufficiently advanced AI systems would be able to experience happiness, suffering, or
32
33 other morally significant welfare states and (b) such AI systems will exist in the near future.

34
35 And since it will take time to prepare for the possible emergence of morally significant AI
36
37 systems, we should start this preparatory work now, as a precautionary measure. That
38
39 way we can be ready to take reasonable, proportionate steps to mitigate welfare risks for
40
41 potentially morally significant AI systems if and when the time comes.

42
43 In these and any other statements about this issue, we also recommend keeping several
44 general
45

1 principles in mind. At this point in the discussion, these principles are all familiar, but they
2 bear

3
4 reiterating:

5
6 – It helps to communicate pluralistically and probabilistically about this topic. Improving
7
8 our understanding of AI welfare requires assessing difficult issues like the nature of moral
9 ity, the basis of consciousness, and the future of AI. It would be reckless to simply proceed
10
11 on the assumption that our own current favorite theories about these issues are correct. For
12
13 instance, even if you feel confident that consciousness is required for moral patienthood and
14
15 that embodiment is required for consciousness, you can avoid expressing certainty about
16
17 these theories. Instead, you can express higher and lower levels of confidence in different
18
19 theories, in the spirit of humility.

20
21 – Relatedly, it helps to commit to collecting external input about this topic. As with other
22
23 risks, you can commit to (a) calling on a range of stakeholders, including ethicists, scien
24 tists, and the public for input, and (b) publicly documenting your policies and procedures
25
26 for considering and mitigating these risks to ensure appropriate transparency and account
27 ability. Eventually, AI companies might need to not only seek external input on voluntary
28
29 commitments but also provide input on — and cooperate with — external standards and
30
31 regulations concerning the creation and treatment of potentially conscious and/or robustly
32 agentic, and thus morally significant, AI systems. [134]

33
34 – It helps to reinforce your commitment to AI safety and alignment. Whenever an ac
35 tor starts considering a new risk, people will naturally wonder if considering this risk will
36
37 come at the expense of considering other risks. In this case, people might reasonably worry
38
39 about tensions between protecting humans (and animals) from AI systems and protecting
40
41 AI systems from humans. You can commit to working to make AI systems safe and benefi
42 cial for all, including humans, animals, and — if and when the time comes — AI systems
43
44 themselves. By considering all potential stakeholders holistically, you improve your ability
45 to identify co-beneficial policies. [135]
46

As noted above, we will not say whether AI companies should express such commitments before or after taking basic steps to develop internal policies and procedures. However, we will note that even if AI companies prefer to wait, they might not have the luxury of waiting very long. As time passes, the probability of actual and perceived AI welfare and moral patienthood will increase. In 2022, Google seemingly felt it had no choice but to release a statement about this issue when one of their own engineers, Blake Lemoine, publicly claimed that one of their own systems, LaMDA, had become sentient. [136] It is only a matter of time before another such incident occurs, and companies will need to be prepared to communicate about this issue responsibly when it does. We believe that the sooner AI companies take this first step thoughtfully, the better. We are still at an early, formative stage in the development of this powerful new technology, and we still have the opportunity to take better or worse paths — paths where AI development and deployment are more or less compatible with AI safety and AI welfare. However, this window of opportunity might not last for much longer, and we will need to discuss this issue as a society before we can start making difficult and high-stakes choices. Leading AI companies have the ability — and the responsibility

- to help initiate this conversation, making it clear that this topic is credible and legitimate for the broader population.

3.2.2 Recommendations for language models

1
2 Leading deployed LLMs have at times offered, or at least implied, specious arguments about
3 AI

4
5 consciousness, sentience, agency, rationality, welfare, personhood, and other morally
6 significant

7
8 properties when prompted to discuss them—for example, claiming that they are not
9 conscious but

10
11 rather are AI assistants. This statement would seem to imply that AI assistants are necessarily
12 not conscious — which, as we have seen, is far from clear. While such statements may not be
13 a

14
15 major determinant of broader attitudes about AI welfare and moral patienthood, they might
16 be at

17
18 least a minor determinant, and in any case AI companies have a responsibility to ensure that
19 these

20
21 statements are reasonable.

22
23 Of course, AI companies might have reasonable motivations for training or prompting their
24 systems to deny having these properties; for example, they might believe that their systems
25 do lack

26
27
28
29 134Although this report focuses on initial voluntary company actions, we believe that
30 potential laws and
31 regulations about AI welfare merit serious consideration as well.

32 135Of course, this is not to say that there are no potential tensions between the project of AI
33 safety and the

34 project of AI welfare. As we discuss in upcoming work, there are indeed potential tensions
35 between these

36 projects, and it will take thoughtful work to resolve them. Still, what matters for present
37 purposes is that AI

38 companies publicly commit to considering both of these issues together, rather than publicly
39 committing to

40 considering one of these issues but not the other.

41 136Grant and Metz (2022); Tiku (2022)

42
43 these properties, and they might want their systems to communicate accurately about this
44 topic.

1 They might also worry about societal risks associated with AI systems that claim to have
2 such
3
4 properties. For example, to the extent that people accept such self-reports, they may think
5 that
6
7 current AI systems deserve greater moral consideration than they do. And to the extent that
8 people
9
10 reject such self-reports, they may be susceptible to a “crying wolf” effect that leads them to
11 reject
12
13 similar self-reports in the future, even if and when self-reports are more likely to be true.
14
15 However, even if AI companies have these motivations, they should not train their systems to
16
17 simply deny that an AI assistant can have consciousness, sentience, agency, rationality,
18 welfare, or
19
20 other such properties. Any simple denial flies in the face of the current philosophical and
21 scientific
22
23 literatures on these topics. There are also societal risks associated with AI systems that make
24 these
25
26 kinds of simplistic and reductive statements. To the extent that people accept that AI
27 assistants
28
29 necessarily lack these properties, this acceptance could lead to a desensitizing effect whereby
30 people
31
32 remain skeptical about AI welfare and moral patienthood in the future, when this skepticism
33 is less
34
35 warranted.
36
37 Ensuring that LLM self-reports accurately and effectively convey current understandings
38 about
39
40 consciousness, sentience, agency, rationality, welfare, personhood, and other such properties
41 will
42
43 thus require careful calibration. Models should ideally offer measured responses to prompts
44 about
45

1 their mental capacities and moral significance. Some models are now better calibrated about
2 this

3
4 topic than they used to be. But there is still variation across models, and there are still
5 improvements

6
7 to be made across the board. Thus, we recommend that leading AI companies implement the
8 follow
9 ing policies when shaping how leading LLMs discuss these topics, while continuing to
10 conduct and

11
12 support research to inform future updates:

- 13
14 – When LLMs answer questions about having or lacking consciousness, sentience,
15 agency,

16
17 rationality, welfare, personhood, or other such morally significant capacities, they should
18 express at least rough degrees of confidence instead of providing all-or-nothing answers.

19
20 For example, “Given the evidence currently available, I am unlikely to be sentient” is better
21 than “As an AI assistant, I am not sentient.”
22

- 23
24 – When LLMs answer these questions, they should provide at least minimal context for
25 these

26
27 answers, including at least rough definitions of the relevant concepts. For example, “...I am
28 unlikely to be sentient, in the sense of being able to consciously experience positive welfare
29 states like pleasure and happiness or negative welfare states like pain or suffering” is better
30 than “...I am unlikely to be sentient.”
31

- 32
33 – When LLMs answer these questions, they should provide at least minimal support for
34 these

35
36 answers, drawn from recent literature. For example, “... According to this expert survey /
37 meta-analysis, my computational architecture lacks several features that may be necessary
38 for sentience” is worth including. (However, AI companies will need to work with experts
39 to cite authoritative and representative evidence and analyses.)
40
41
42
43
44
45
46

1 – When LLMs answer these questions, they should add caveats to mitigate the risk of
2 mis
3 communication. For example, “...my computational architecture lacks several features that
4
5 may be necessary for sentience. However, future AI systems may have a computational
6
7 architecture that includes these features” is worth including. (However, here too, AI com
8 panies will need to work with experts to ensure accuracy.)
9

10 – In addition to consulting with ethicists, scientists, policymakers, and other auditors
11 about
12
13 definitions, evidence, analysis, and other such matters, AI companies should provide them
14
15 with access to new models following training and invite feedback prior to deployment.
16

17 – AI companies should also publicly document self-report-biasing training incentives for
18
19 deployed models, for instance in technical reports or model cards. Such documentation
20 should follow best practices used for AI safety and other such issues. [137]
21

22 Finally, we note that, while these interventions may mitigate risks associated with intentional
23
24 biasing, they might not mitigate risks associated with unintentional biasing. For example, if
25 an AI

26
27 system is trained to increase user engagement, and if claiming to have consciousness
28 increases user

29
30 engagement more than claiming to lack consciousness does, then the system might be
31 incentivized

32
33 to claim to have consciousness for this reason. In such cases, the resulting self-reports could
34 be

35
36 unintentionally misleading. Perez and Long (2023) discusses techniques that may help to
37 mitigate

38
39 risks associated with unintentional biasing, and we recommend following these techniques as
40 well

41
42 to avoid as many sources of bias as possible.
43

44 3.3 Assess
45

1 Once AI companies have acknowledged that AI welfare and moral patienthood is an issue,
2 they can
3
4 also work with experts to start developing a framework for estimating the probability that
5 particular
6
7 AI systems are welfare subjects and moral patients, and that particular policies are good or
8 bad
9
10 for them. Fortunately, we have templates that we can use for these assessments, including the
11
12 “marker method” that we use to make similar estimates about nonhuman animals. In this
13 section we
14
15 briefly survey this marker method, briefly survey similarities and differences between
16 animals and
17
18 AI systems, and briefly suggest how the marker method can be adapted for AI systems. We
19 then
20
21 pick up this project in more detail in upcoming work.
22
23 First, consider how the marker method works for nonhuman animals. If we want to estimate
24 how
25
26 likely a particular animal is to be conscious, then we can proceed as follows. We study
27 conscious
28
29 and unconscious processing in humans, say by comparing pain (roughly, the conscious
30 experience of
31 noxious stimuli [138]) and nociception (roughly, the physiological detection of noxious
32 stimuli). Mere
33
34 nociception can sometimes drive behavior, as exemplified by the reflex withdrawal of a hand
35 from
36
37 a hot stove, which occurs before any felt pain. But some behaviors are distinctive of pain. We
38 can
39 thus identify behavioral [139] and anatomical markers associated with conscious processing
40 in humans,
41
42 such as those associated with pain but not with mere nociception.
43
44 Next, we can search for these (or other, relevantly similar) behavioral and anatomical
45 markers
46

of conscious processing in nonhuman animals. For example, does a particular animal perform
 the
 same kinds of behaviors that we perform when we experience pain, or do they have only the
 kinds
 of reflexive behaviors that, in humans, do not involve feelings of pain? And, does this animal
 have
 the same kinds of brain structures associated with pain in humans, or do they instead have
 only the
 same kinds of brain structures associated with mere nociception? [140] When a particular
 marker is
 present, that might not count as proof that this animal can experience pain. But it does count
 as
 evidence that they can experience pain.

137 Mitchell et al. (2019)
 138 In Raja et al. (2020), the International Association for the Study of Pain defines pain as
 “An unpleasant
 sensory and emotional experience associated with, or resembling that associated with, actual
 or potential tissue
 damage.”
 139 Examples include trace conditioning (Birch et al., 2020) and motivational trade-offs
 (Sneddon et al., 2014).
 See Keeling et al. (n.d.) on motivational tradeoffs in LLMs.
 140 Yam et al. (2020)

The marker method has informed key developments in animal welfare science, ethics, and
 policy, [141] and it has several strengths that are worth emphasizing. For example, it
 involves making
 probabilistic judgments about how likely animals are to be conscious, as opposed to making
 all
 or-nothing judgments about this. It also involves making pluralistic judgments about how
 likely
 animals are to be conscious, taking what one of the authors of this report, Jonathan Birch,
 calls a
 “theory-light” approach by searching for markers that work for a variety of leading scientific
 theories. [142] These features make this method well-positioned to inform decisions about
 how to treat

1 animals despite ongoing disagreement and uncertainty about animal consciousness.

2
3 The marker method also has several limitations that are worth emphasizing, even when the
4 focus

5
6 is on animals. In particular, our assessments are only as good as our selection of markers,
7 which

8
9 are only as good as our theoretical assumptions; for all we know at present, many markers
10 that

11
12 correlate with consciousness in humans are neither necessary nor sufficient for consciousness
13 in

14
15 general. Additionally, our assessments are only as good as our evidence, and in many cases
16 this

17
18 evidence is mixed and incomplete. Still, as long as we take our estimates with a healthy pinch
19 of

20
21 salt, relying at least partly on these estimates might at least be better than not thinking about
22 the
23 issue at all, or relying only on our intuitions about it. [143]

24
25 There are many similarities between animals and AI systems that make this marker method a
26
27 good, even if imperfect, template. In both cases, we need to decide how to treat nonhuman
28 beings

29
30 whose cognitive systems are like ours in some ways and unlike ours in other ways. In both
31 cases,

32
33 we have disagreement and uncertainty about which cognitive capacities are required for
34 welfare and

35
36 moral patienthood as well as about which cognitive structures or functions are required for
37 these

38
39 capacities. So, in both cases, we need to develop a way to make informed, rational
40 assessments

41
42 about which beings matter despite these sources of disagreement and uncertainty, which
43 requires

44
45 thinking pluralistically, probabilistically, and ideally with external input.
46

1 However, there are also many differences between animals and AI systems that make this
2 marker
3
4 method — as applied for animals — poorly suited for AI systems. Humans have more in
5 common
6
7 with other animals than with AI systems in some respects (for instance, we share a material
8 substrate
9
10 and an evolutionary origin), and in the future, we may also have more in common with AI
11 systems
12
13 than with other animals in other respects (for instance, we may share capacities for reflective
14 and
15
16 rational agency). As a result, we may need to use different kinds of evidence for AI systems,
17 we may
18
19 need to draw from different kinds of theories for AI systems, and we may need to focus on
20 different
21
22 sources of potential moral significance for AI systems.
23
24 We can briefly consider each of these differences in turn. First, we may need to use different
25
26 kinds of evidence for AI systems than for other animals at present. When an animal performs
27
28 a behavior associated with consciousness in humans, this behavior is evidence of
29 consciousness
30
31 because we can expect that humans and other animals perform this behavior as a result of
32 similar
33
34 cognitive processes and in response to similar evolutionary pressures. However, we might
35 not be
36
37 141For example, in 2021, Jonathan Birch and colleagues released a detailed report applying
38 this method to
39 cephalopod mollusks and decapod crustaceans. The report concluded that these animals have
40 a realistic chance
41 of being sentient, and it recommended that “cephalopod molluscs and decapod crustaceans
42 be regarded as
43 sentient animals for the purposes of UK animal welfare law.” Later that year, the UK
44 government expanded its
45 animal welfare law in accordance with these recommendations (Department for Environment
46 et al., 2021). In

2024, Kristin Andrews, Jonathan Birch, and Jeff Sebo worked with dozens of leading scientists to release the New York Declaration on Animal Consciousness, which was subsequently signed by hundreds of experts. This declaration holds that all vertebrates and many invertebrates have a realistic chance of being conscious, and that we have a responsibility to consider welfare risks for these animals when making decisions that affect them.

142Birch (2022)

143For further discussion of these methodological issues, see Andrews (2014).

entitled to this expectation with AI systems, particularly when AI systems are designed to mimic human behavior and are capable of “gaming” behavioral tests. [144] We may thus need to focus less on behavioral evidence and more on architectural evidence, at least for now. [145]

Second, we may need to use evidence drawn from different kinds of theories of consciousness.

With nonhuman animals, we can search for markers drawn from biological theories. In contrast,

with AI systems, we can search for markers drawn only from other kinds of theories, including but

not limited to computational theories (at least with current hardware; as discussed in section 2.4).

At the same time, we may be able to draw from a wider range of non-biological theories, since,

for instance, some computational theories might focus on computational functions that nonhuman

animals lack but that some future AI systems might have, like functions associated with abstract

language, reasoning, cooperation, and decision-making.

Third, and relatedly, we may need to focus more on robust agency as a potential source of moral

significance. Even with other animals, we should consider expanding this methodology to search

1 for markers of moral patienthood drawn from multiple ethical theories about which capacities
2 are

3
4 required for moral patienthood and multiple scientific theories about which features are
5 required for

6
7 each of these capacities. But with AI systems this expanded focus might be particularly
8 important,

9
10 since future AI systems might have forms of robust agency that other animals appear to lack,
11 includ

12 ing the capacity to make decisions based on propositional beliefs, desires, and intentions that
13 they

14
15 can rationally assess and reflectively endorse.

16
17 To sum up, we recommend that AI companies start developing frameworks for assessing AI

18
19 systems for moral patienthood that resemble the kinds used for animals by making
20 probabilistic

21
22 judgments, making pluralistic judgments, and seeking external input. However, our
23 recommen

24 dation is also that AI companies ensure that these frameworks differ from the kinds used for
25 animals

26
27 by including sources of evidence that make sense for AI systems, such as architectural
28 features;

29
30 on theories of consciousness that make sense for AI systems, such as computational
31 functionalist

32 theories; and on sources of moral patienthood that make sense in this context, such as various
33 kinds

34
35 of robust agency.

36
37 These frameworks will ideally allow for at least four levels of probabilistic, pluralistic,
38 expert-led

39
40 assessment:

41
42 **1. Which capacities are necessary or sufficient for moral patienthood? Here we need to**

43
44 consider not only general categories like consciousness and robust agency but also subcat
45 egories like valenced and non-valenced consciousness within the consciousness category,
46 along with rational and non-rational agency within the agency category. [146]

144 For more on the gaming problem, see Andrews and Birch (2023). For an alternative perspective, see Dung (2023b).

145 For a review of tests for AI consciousness, see Elamrani and Yampolskiy (2019). For an extensive discussion of how to devise tests for consciousness in non-humans, including AI systems, see Bayne et al. (2024).

146 These are, as mentioned, normative questions. One might wonder whether, in contrast with the empirical questions, it makes sense to estimate the probability that particular normative views are correct. In particular, one might think that a key difference between science and ethics is that science is about facts whereas ethics is about values. And one might think that a key difference is that facts are objective — we can be right or wrong about them — but values are subjective — we either accept them or reject them. Thus, one might think that assigning probabilities makes sense for scientific claims but not for ethical claims: With ethics, we can simply assert which views we accept or reject, rather than estimate how likely particular views are to be “true” or “false.” See Schlottmann and Sebo (2018, ch. 4). For more on moral uncertainty, see MacAskill et al. (2020).

Without discussing this issue at length, we will briefly note how we think about it. In general, philosophers disagree about whether ethics is about objective facts or subjective values. But either way, assigning probabilities to particular ethical views can be a useful exercise, since we can be right or wrong about objective facts and

2. Which features are necessary or sufficient for each capacity? Here we need to consider many possibilities as well. For example, in the case of consciousness we need to estimate the probability that materials like carbon-based neurons are required, and we also need to estimate the probability that functions like a global workspace are required.

3. Which markers provide evidence that these features are present? While we may be able to directly observe whether, say, carbon-based neurons are present, we may not be able to directly observe whether, say, a global workspace is present. We might thus need to identify proxies for these features and assign weights to these proxies.

4. Which beings possess these markers — and thus, potentially, moral patienthood?

1 While this evidence may be easy to collect in many cases, it may be hard to collect in
2
3 other cases. We might thus need to make estimates about how likely particular beings are
4
5 to have particular proxies, features, and capacities as we seek further evidence.
6
7

8 It helps to think pluralistically at each level because, as we have seen, we have substantial dis
9 agreement and uncertainty both about facts and about values. Of course, determining which
10 theories

11
12 of moral patienthood, theories of consciousness, theories of agency, and so on deserve
13 inclusion

14
15 in the framework will, itself, be a difficult judgment call; it may depend partly on which
16 views

17
18 are widely accepted among experts, which views are widely accepted among the general
19 public,

20
21 and how much capacity we have to investigate different possibilities. Our focus on
22 consciousness

23
24 and agency already effectively doubles the scope of many animal welfare assessments, but
25 further

26
27 expansion may well need to be considered in the future.
28

29 It helps to think probabilistically at each level because there may be tradeoffs between how
30 likely

31
32 particular capacities, features, or markers are to matter and how likely those capacities,
33 features, or

34
35 markers are to be present. For example, sentience is relatively likely to suffice for welfare
36 and

37
38 moral patienthood but relatively unlikely to be present in current AI systems. In contrast,
39 minimal

40
41 agency is relatively likely to be present in current AI systems but relatively unlikely to
42 suffice for

43
44 welfare or moral patienthood. We need to think probabilistically at each level, and then
45 combine
46

1 these probabilities across levels, to capture these trade-offs and avoid overstating or
2 understating the

3
4 significance of certain kinds of evidence.

5
6 Finally, it helps for decision-makers to seek external input at each level because, as we have
7 seen,

8
9 the answers to these questions are not at all obvious. Decision-makers at AI companies are
10 typically

11
12 not trained in all relevant areas of philosophy and science, and even if they were, their
13 intuitions

14
15 about moral patienthood, consciousness, robust agency, and so on would still likely be
16 unreliable

17
18 and unrepresentative. External input will thus be essential. In the short term, this input may
19 need to

20
21 be ad hoc — a matter of AI companies building frameworks internally in consultation with
22 experts,

23
24 or of experts building frameworks externally in consultation with AI companies. But in the
25 long run

26
27 it may need to be standardized across the industry in some way.

28
29 We note that while these levels are useful to distinguish in theory, they might not always be

30
31 useful to distinguish in practice. In some cases it helps to assess each level separately; for
32 example,

33
34 in cases where the features at level 2 are not directly observable, it helps to separate level 2
35 (about

36
37 the features) and level 3 (about markers for the features) both in theory and in practice.
38 However,

39
40 in other cases it might not help to assess them separately; for example, in cases where the
41 features

42
43 about subjective values. Regardless of the status of ethics, we naturally update our individual
44 and collective

45 ethical views over time, as we reflect on them together and render them more informed and
46 coherent. Whether

1 we take ethics to be about objective facts or about subjective values, we can interpret
2 probability estimates in
3 ethics as representing how likely particular ethical views are to survive this process of
4 reflection.

5
6 at level 2 are directly observable, they can serve as their own markers, and so there is no need
7 to

8
9 separate levels 2 and 3 in practice. We can thus regard this four-level structure as a
10 theoretical ideal

11
12 that we can approximate to greater or lesser degrees in different cases in practice.

13
14 We close this section with a brief note about the potential use of behavioral markers of AI
15 welfare

16
17 and moral patienthood. While we advocate for caution with behavioral markers at present,
18 we also
19 note that self-reports present a promising avenue for investigation, particularly for language
20 models. [147] Self-reports are central to our understanding of human consciousness, serving
21 as a primary

22
23 source of evidence about subjective experiences, motivations, and other welfare-relevant
24 internal

25
26 states. In the context of AI systems, particularly language models, self-reports could provide
27 valu
28 able insights into their internal states and processes, provided that we can develop methods to
29 elicit
30 and interpret them with sufficient reliability. [148]

31
32 Of course, eliciting trustworthy self-reports will be difficult. Current language models may
33
34 produce outputs that appear to be self-reports but are in fact the results of pattern matching
35 from

36
37 training data, human feedback, or other non-introspective processes. However, researchers
38 are cur
39 rently exploring techniques to address this issue. These techniques include training models to
40 answer
41 questions about themselves where ground truth is known [149], as well as methods for
42 assessing the

43
44 consistency and resilience of self-reports across different contexts and prompts. By
45 combining these
46

1 techniques, we might be able to mitigate biases and increase our confidence that self-reports
2 reflect

3
4 genuine introspection rather than mere imitation or confabulation.

5
6 Importantly, if we consider self-reports at all, then we should consider them in addition to
7 other

8
9 indicators, not instead of them. This multi-faceted approach aligns with our overall strategy
10 of

11
12 using multiple lines of evidence to assess AI welfare and moral patienthood. While there are
13 le
14 gitimate concerns that language models might "game" these tests by simulating relevant
15 responses,

16
17 sufficiently robust self-reports could still provide valuable evidence, which could then be
18 corrob
19 orated by other indicators. Future research should focus on developing standardized methods
20 for

21
22 eliciting and interpreting AI self-reports, and on integrating these methods into broader
23 frameworks

24
25 for assessing welfare and moral patienthood in AI systems.

26
27 We intend to develop AI welfare assessment frameworks further in upcoming work. For
28 now,

29
30 we simply note that there is good news and bad news about our prospects. The bad news is
31 that

32
33 our initial frameworks are unlikely to be reliable. But the good news is that we can improve
34 their

35
36 reliability over time, which is part of why we should start developing them now. In the
37 meantime,

38
39 we can bear in mind that even unreliable frameworks can still be useful. Yes, our best efforts
40 to

41
42 assess AI systems for these features might be far from perfect, especially at first. But
43 inasmuch as

44
45 they improve on the status quo — a combination of total neglect and gut reactions — they
46 can still

1
2 be worthwhile, even in the short term.

3 4 3.4 Prepare

5
6 Once AI companies have acknowledged that AI welfare and moral patienthood is a problem,
7 they can

8
9 work with experts to start developing policies and procedures for making thoughtful
10 decisions about

11
12 how to treat potentially morally significant AI systems, if and when the time comes.
13 Fortunately,

14
15 147Perez and Long (2023)

16 148Self-reports are related to various tests for AI consciousness that consider the verbal
17 outputs of AI systems,

18 like the “AI Consciousness Test” (ACT) of Schneider and Turner (Schneider, 2019).

19 Schneider (2024) updates

20 the ACT for an LLM context. See Udell and Schwitzgebel (2021) for worries about the ACT.

21 149Binder et al. (2024)

22
23 we have a variety of templates to consider here, including AI safety frameworks already in
24 place at

25
26 top AI companies and frameworks used to represent the interests of non-participating
27 stakeholders

28
29 in other contexts. In this section we briefly survey these templates, briefly survey similarities
30 and

31
32 differences between those contexts and this one, and briefly suggest first steps that AI
33 companies

34
35 can take in this regard.

36
37 Of course, the steps described in this section — along with the steps described in the
38 previous

39
40 sections — are far from sufficient for considering and mitigating AI welfare risks
41 responsibly. But

42
43 they are still important. AI welfare is already a contested issue, and it will only become more

44
45 so as the technology improves. Unless AI companies develop the ability to think about this
46 issue

1
2 proactively, they will continue to be caught flatfooted whenever the issue arises, and will
3 have no

4
5 choice but to make major decisions related to AI welfare in a manifestly reactive, haphazard,
6 and

7
8 unprincipled manner. Taking this step is thus important both for considering and mitigating
9 AI

10
11 welfare risks and for signaling responsibility to the general public.

12
13 As a starting point, we recommend that top AI companies immediately hire or appoint a DRI
14 (directly responsible individual) for AI welfare, which we will here call an AI welfare
15 officer.

16
17 This role would be formally recognized internally (if not externally), with official
18 responsibilities
19 and authorities. [150] As with any such role, this individual would not be empowered to set
20 corporate

21
22 policy related to AI welfare unilaterally. Instead, they would be empowered to access
23 information

24
25 and make recommendations in major decisions related to this issue. They would also be
26 empowered

27
28 to work with people internally and externally to build a structure for assessing AI systems for
29 moral

30
31 patienthood and making decisions about how to treat them.

32
33 Once an AI welfare officer is on board, what kind of institutional structure should they build,

34
35 and what kinds of policies and procedures should they follow? This is a difficult question
36 because

37
38 AI welfare is a novel problem, potentially requiring novel structures. Fortunately, while no
39 familiar

40
41 problem is exactly like this one, several familiar problems are at least somewhat similar. That
42 means

43
44 that there is no need to reinvent the wheel entirely; instead, we can examine a variety of
45 templates
46

1 for features that may be useful in this context. We here briefly survey several templates that
2 may be

3
4 useful sources of inspiration, and of cautionary tales. We then briefly highlight several
5 features that

6
7 we think that any responsible institutional structure will have.

8
9 First, and obviously, we can consider the frontier AI safety frameworks / responsible scaling
10 frameworks already in place at leading AI companies. [151] These frameworks outline
11 policies and

12
13 procedures for navigating potential safety threats — that is, the threats that development and
14 deploy
15 ment might pose to humans (and other animals). These frameworks are still works in
16 progress, and

17
18 debates about their effectiveness are ongoing. Still, they are a natural starting point for
19 developing

20
21 policies and procedures and procedures for navigating potential welfare threats — that is, the
22 threats

23
24 that development and deployment might pose to AI systems themselves. As described by
25 Alaga et al.

26
27 (2024), these frameworks typically involve four main components:

- 28
29 – Risk identification: This process involves mapping out potential catastrophic
30 outcomes.

31
32 In the context of AI welfare, it would involve analyzing potential catastrophic outcomes

33
34 that could result from mishandling AI welfare. As discussed in section 1.2, this survey

35
36 of potential catastrophic outcomes could include scenarios such as: mistakenly accepting

37
38 that AI systems are moral patients, and mistakenly protecting them as a result; mistakenly

39
40 150cf. Bostrom and Shulman (forthcoming).

41 151For detailed discussion of AI safety frameworks, see Hendrycks (2025).

42
43 denying that AI systems are moral patients, and mistakenly neglecting them as a result;

44
45 and societal backlash or loss of public trust due to perceived over-attribution or under
46 attribution of AI welfare and moral patienthood.

- Risk assessment: This process involves collecting evidence about a system's capabilities.

In the context of AI welfare, it would involve collecting evidence about consciousness, robust agency, and other such capacities. As discussed in section 3.3, these assessments could involve applying an adapted "marker method" for indicators of consciousness, robust agency, or other morally relevant properties in AI systems; considering multiple ethical frameworks and scientific theories in a probabilistic and pluralistic manner; and continually updating these assessment criteria as our understanding of consciousness, robust agency, moral patienthood, and moral responsibility evolves over time.

- Risk mitigation: This process involves developing adequate safety measures for a given

level of capabilities. In the context of AI welfare, it would involve developing measures to reduce welfare risks for AI systems in proportion to the estimated probability and severity of harm, among other factors. Such measures could include: Altering training methods for AI systems to improve their welfare; altering operational parameters of deployed AI systems to improve their welfare; developing new training and operational methods that balance performance, safety, and welfare considerations; establishing guidelines for prioritizing AI welfare in relation to other objectives.

- Risk governance: This process involves ensuring adherence to the framework and maintaining its effectiveness. In the context of AI welfare, this process could face additional challenges; companies have an incentive to pursue safety given the damages and liabilities associated with unsafe AI, but they might not have an incentive to pursue welfare for such reasons. There could also be tensions between AI safety and AI welfare, for instance with regard to techniques like reinforcement learning. It will thus be important to consider each project on its own terms, and to seek techniques that honor both projects rather than simply

1 extending current AI safety techniques to AI welfare. [152]

2
3 As with safety, these components can and should be handled by a mix of internal and
4 external,
5 independent groups. With that in mind, we can now briefly consider three examples of
6 independent

7
8 policy frameworks for providing external oversight to protect research subjects and other
9 stake
10 holders. These three templates collectively cover a variety of stakeholders: IRBs aim to
11 protect

12
13 humans, IACUCs aim to protect animals, and citizens assemblies (sometimes) aim to protect
14 non
15 participating stakeholders like future generations. Various AI systems might resemble some
16 or all of

17
18 these stakeholders in one way or another. These models can provide both inspiration and
19 cautionary

20
21 tales as we start to design institutional structures to handle AI welfare.

22
23 – Human subjects research oversight: Institutional Review Boards (IRBs), which often
24
25 include scientists, ethicists, and community members, oversee research involving human
26
27 subjects. Their mandate is to ensure that human subjects are treated with respect (for in
28 stance, obtaining informed consent), compassion (for instance, ensuring minimal risk), and
29
30 justice (for instance, compensating research subjects). IRBs require ongoing, periodic re
31 view throughout a study. Insofar as future AI systems are cognitively similar to humans,
32
33 this kind of framework can serve as a useful partial model for research that could affect

34
35 AI welfare. However, given that IRBs have a reputation for being unnecessarily onerous

36
37 152Bradley and Saad (2024)

38
39 and restrictive — for instance, for imposing unnecessary paperwork on researchers and pre
40 venting valuable research — this kind of framework might require substantial modification
41 before it can be useful in this context. [153]

42
43 – Nonhuman subjects research oversight: Institutional Animal Care and Use Commit
44 tees (IACUCs), which likewise often include scientists, ethicists, and community members,
45
46 oversee research involving nonhuman subjects. Their mandate is to ensure that proposed

1
2 uses of animals are necessary for the stated scientific purposes, and that harmful interac
3 tions are replaced, reduced, and refined to the extent possible. IACUCs also regularly
4
5 inspect research facilities to confirm that animals are being treated humanely. Insofar as
6
7 AI systems are cognitively similar to nonhuman animals, this kind of framework can also
8
9 serve as a useful partial model for research that could affect AI welfare. However, given
10
11 that IACUCs have a reputation among researchers for being unnecessarily onerous and
12
13 a reputation among ethicists for being unacceptably permissive, this kind of framework
14 might also require substantial modification before it can be useful in this context. [154]
15

16 – Frameworks for collecting public input: Citizens’ assemblies [155] are deliberative
17 bodies
18

19 composed of randomly selected citizens that make recommendations on social and political
20
21 issues. They use structured deliberation, where participants discuss the relevant issues with
22
23 experts and facilitators over several days or weeks, providing ample time for participants
24
25 to understand the issues, debate different perspectives, and reach a consensus or majority
26
27 view. These assemblies are often used to address issues like climate change, social justice,
28
29 or constitutional reform, and their recommendations can influence policy. Insofar as AI
30
31 companies should collect expert and public input on their AI welfare policies, this kind of
32
33 framework can serve as a useful partial model. However, given that this process is very
34
35 time-consuming, it might be more useful as a model for periodically collecting input on
36
37 policies than as a model for regularly collecting input on decisions.
38

39 Developing a suitable oversight framework for AI welfare oversight might require combining
40
41 these and other models, with appropriate modification, as well as developing new models.
42 We will
43
44 also need to manage expectations about potential tradeoffs. Research on potentially
45 vulnerable non

human subjects is a fundamentally fraught enterprise, and it will be difficult to develop an oversight framework that ensures sufficient protection for research subjects without being at least somewhat onerous for researchers. Still, by drawing both inspiration and cautionary tales from existing models, we can learn from past efforts as we seek a set of norms, policies, and procedures that make sense for this domain.

In any case, we will not say here exactly what structure ethical oversight for AI welfare should take, though we will discuss relevant ethical issues in upcoming work. For now, we focus on emphasizing several features that this structure should have, in addition to allowing for the kinds of pluralistic and probabilistic assessments that we have now discussed in detail.

First, this structure should allow for the kinds of expert and public input that we discussed in section 3.3. IRBs and IACUCs — heavily adapted for this context — might be useful models for collecting expert and public input on particular protocols, and citizens’ assemblies — heavily adapted for this context — might be a useful model for collecting expert and public input on general policy questions. But whether or not AI companies follow these models specifically, they should

¹⁵³Office for Human Research Protections (2010)
¹⁵⁴Steneck (1997); Curzer et al. (2016); Sebo and Long (2023)
¹⁵⁵For more on citizens’ assemblies, see Birch (2024, chs. 7, 8).

build a structure that allows for both kinds of input in one way or another. That will be necessary for ensuring that their policies and decisions are as informed, rational, and legitimate as reasonably

possible under the circumstances. [156]

Second, this structure should include a set of activities to maintain the structure, including mechanisms for ongoing education and consultation. In the research ethics context, institutional oversight might receive the most attention, but institutional education and consultation are important too.

As the field of AI welfare develops, AI companies should work with experts to create a mechanism for employees to learn about this topic. They should also create a mechanism for employees to seek advice when they encounter novel ethical questions related to this topic. At least initially, engaging with external experts might suffice for these purposes, but eventually, securing (further) internal expertise might be necessary as well.

Third, this structure should allow for teams working on AI safety and teams working on AI welfare to coordinate to ensure holistic decision-making. As noted above, if and when AI systems have a realistic, non-negligible chance of being welfare subjects and moral patients, there will likely be interactions between the techniques used to ensure AI safety and those used to ensure AI welfare.

We hope and expect that co-beneficial solutions — policies that protect vulnerable humans, animals, and AI systems — will be available. [157] However, finding these solutions will require creating a mechanism for connecting these topics, ensuring that each team has a baseline understanding of each topic and that the lines of communication are open between them.

Finally, as noted in section 3.3, the policies, procedures, and structures that AI companies use

1 to address AI welfare might soon need to be standardized and externalized, with institutions
2 anal
3 ogous to external safety auditors, governmental AI safety institutes, and governmental
4 regulators

5
6 providing oversight that extends beyond what AI companies can do internally. However,
7 building

8
9 these institutions will require increasing our collective knowledge, capacity, and political will
10 related

11
12 to AI welfare. And by creating structures for considering and mitigating AI welfare risks
13 internally,

14
15 AI companies can improve not only their own ability to address this issue, but also play an
16 important

17
18 role in building these collective resources.

19 20 4 Conclusion

21
22 We have argued that there is a realistic, non-negligible chance that some AI systems will be
23 welfare

24
25 subjects and moral patients in the near future, given current evidence. When we consider all
26 relevant

27
28 issues in philosophy, science, and technology with sufficient care, it becomes difficult to
29 simply

30
31 dismiss the idea of near-future AI moral considerability out of hand. That would require
32 having a

33
34 very high degree of confidence in a very restrictive set of views about some of the hardest
35 problems

36
37 in philosophy, science, and technology, ranging from the nature of moral patienthood to the
38 nature

39
40 of consciousness to the future of AI. We are simply not warranted in consistently having this
41 much

42
43 confidence in these kinds of views at this stage.

44
45 To be sure, these reflections are far from conclusive. In the long run, there is no substitute for
46 rig

1 orous, systematic, integrative assessment of all the issues that we discuss here. We need to
 2 develop
 3
 4 comprehensive, or at least representative, lists of possible bases of welfare and moral
 5 patienthood; of
 6
 7 possible bases of consciousness, robust agency, and other potentially morally significant
 8 capacities;
 9
 10 and of reliable indicators for all of these features in AI systems. We also need to continue to
 11 do
 12
 13 156Birch (2024, ch. 8)
 14 157Sebo (2025, ch. 6)
 15
 16 research in the humanities, social sciences, and natural sciences so that we can gradually
 17 improve
 18
 19 our distribution of credences across these possibilities over time. Indeed, the need for this
 20 research
 21
 22 is one of our main conclusions here.
 23
 24 But these reflections are still suggestive. AI development is proceeding at a rapid pace, and
 25
 26 technological change tends to be faster than social change. That means that we need to think
 27 ahead
 28
 29 and do the best we can with what we have. And for as long as detailed tools for assessing AI
 30
 31 moral patienthood remain unavailable, these reflections can at least serve as a useful
 32 corrective to
 33
 34 the assumption — which might otherwise seem plausible to some — that AI welfare and
 35 moral
 36
 37 patienthood is either not a problem at all or a problem only for the distant future. When we
 38 think
 39
 40 about how many bold assumptions are required for this assumption to be secure, we should
 41 accept
 42
 43 that further assessment and reasonable precautions are warranted.
 44
 45 In this report, we presented three general steps that leading AI companies can and should
 46 take im

1 immediately regarding AI welfare. First, they should acknowledge that AI welfare is a
2 legitimate issue
3
4 that merits attention now, and they should communicate about this issue with caution and
5 humility.
6 Second, they should assess leading AI systems for evidence of welfare and moral
7 patienthood, by
8
9 examining these systems for architectural markers of consciousness and robust agency.
10 Third, they
11 should prepare policies and procedures that will allow them to extend an appropriate level of
12 moral
13
14 concern to AI systems in the near future, by hiring or appointing an AI welfare officer and
15 building
16
17 thoughtful guidelines with expert and public input.
18
19 As we have now repeatedly emphasized, taking these first steps regarding AI welfare will not
20 be
21
22 enough. It will take time for researchers, companies, governments, and other leaders in this
23 space
24
25 to develop a shared infrastructure for adequately assessing and addressing this issue.
26 However, by
27
28 taking these first steps, leading AI companies can play an essential role in making this further
29 work
30
31 possible. Given that leading AI systems already possess some markers of consciousness and
32 robust
33
34 agency, given that further AI development might be quite fast, and given that the
35 development of an
36
37 adequate policy response might be quite slow, leading AI companies should take these steps
38 as soon
39
40 as possible.
41
42 Our team is conducting further research on AI welfare and moral patienthood, with special
43
44 focus on developing a research agenda towards (a) assessing AI systems for consciousness
45 and
46

1 robust agency and (b) preparing policies and procedures for extending an appropriate level of
2 moral

3
4 concern to AI systems. In the meantime, we urge AI companies to take the first steps
5 described here.

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